transcribed DNA (e.g., ribosomal RNA, histones), and diverged copies of ancestral genes (e.g., globin family genes).

19. The most direct method of determining the direction of transcription of the histone genes would be to clone and sequence the region, from which transcriptional information can be ascertained.

21. Cloning and then sequencing the region would provide the answer. Analysis would show genes of similar sequence to the active genes but lacking the sequences for transcription.

23. It is very rich in A-T sequences. Density is proportional to G-C content. Since the molecule has a low density, it has low G-C, and therefore high A-T.

25. Spaces between the nucleosomes must contain many promoter sequences. For the DNA to be digested, it must be unprotected. Since we see little transcription, the promoters must be missing and must have been destroyed by the nucleases.

27. Highly repetitive DNA must be located in these regions. Since most highly repetitive DNA is not transcribed, the results suggest that centromeric and telomeric regions are not transcribed.

29. The C-value paradox involves the issues of the excessive amounts of DNA in eukaryotic cells and the difference between eukaryotic species that seem to have similar complexity. It is explained by the large amount of structural DNA in chromosomes as well as the large amounts of short and long interspersed elements (SINEs and LINEs).

Critical Thinking Question:

1. Comparative DNA studies can be helpful in understanding the roles of the various types of DNA in the eukaryotic chromosomes if there are cases in which there are remarkably large differences in the amount of DNA in similar species. It can then be inferred that the basic developmental plan of an organism is contained in the one with the lower amount of DNA, and the extra DNA in the species with more DNA may be superfluous. We do have cases in which amphibians differ by as much as one hundred times the amount of DNA found in similar species. The puffer fish has only one-sixth the amount of DNA as other higher eukaryotes.

Chapter 16 Gene Expression: Control in Eukaryotes

1. See figure 16.5.

3. Since 5-azacytidine prevents methylation, the observed increase in transcription suggests that the presence of methyl groups inhibits transcription.

5. Genomic equivalence means that all of the cells of a multicellular eukaryotic organism are genetically identical. Yet, cells in different tissues and different regions of the organism are phenotypically different, expressing different suites of genes. Explaining differences in gene expression among cells that are genetically identical is a major question of eukaryotic genetics.

7. The three classes of segmentation genes in *Drosophila* are gap, pair-rule, and segment polarity. Mutations in gap genes leave gaps of missing segments. Mutations in pair-rule genes leave gaps of even or odd sets of segments. Mutations of segment polarity genes cause changes in all segments, generally the change in anterior or posterior portions of each.

9. In the development of the early *Drosophila* embryo, a syncitial blastoderm stage is achieved after thirteen cell divisions. The nuclei are near the surface of the embryo but not surrounded by cell membranes. Thereafter, membranes form, creating a cellular blastoderm.

11. Maternal-effect genes determine four regions of the developing embryo (major gene in parentheses): anterior (*bicoid*), posterior (*nanos*), dorso-ventral (*Toll*), and terminal (*torso*).

13. The helix-turn-helix motif (see box 16.1, fig. 1) consists of two alpha helices separated by a short turn within the protein, providing the structure to interact with DNA. Two other motifs are the zinc finger and the leucine zipper. Another motif, a combination of helix-turn-helix and leucine zipper, is shown in box 16.1, figure 4.

15. Amphibians have very large eggs, development is external to the female, a ready supply of zygotes is available, and they are easily manipulated experimentally.

17. Assuming that each cancer might be controlled by a single locus and assuming that breast cancer appears only in women and prostate cancer appears only in men (sex limited), pancreatic and prostate cancer are probably controlled by autosomal recessive genes; colon cancer is probably controlled by a dominant gene (autosomal or sex linked); and breast cancer by a recessive gene, either autosomal or sex linked.

19. The protein product of the retinoblastoma gene, p105, binds with oncogene proteins. Thus, the protein may somehow suppress transformation; when bound by oncogene proteins, p105 may be rendered ineffective. Hence, p105 seems to act to suppress transformation and thus the gene is called an anti-oncogene.

21. Animal viruses can have DNA or RNA, either single- or double-stranded. They can be enveloped or nonenveloped. They can have simple or complex protein coats.

23. The following are translation mechanisms: normal translation; read-through translation; and splice and then translation.

25. The v and c refer to viral and cellular, respectively. A proto-oncogene is a cellular oncogene within a nontransformed cell.

27. The v-*src* gene has no introns and the virus can function without the gene.

29. We see one band that is common to both cell lines; this band must represent the normal oncogene. The fact that this band is present in both lines indicates that the insertion of a virus has occurred in only one of the two copies of the gene present in the clone 1 cell. If it had inserted within both genes, we should not have seen the normal band. We see a larger fragment in clone 1, indicating that the DNA of the virus does not contain a site for the restriction enzyme used and that the virus has inserted within the restriction sites that define the band, lengthening the region probed. Alternatively, the virus could contain a restriction site and has still inserted in such a way as to lengthen the band probed by inserting between the sequence probed and the original restriction site.

31. The components of an immunoglobulin light chain are V, J, and C regions; the components of an immunoglobulin heavy chain are V, D, J, and C regions.

33. The V-J joining recognition signal is a heptamer and nonamer separated by twenty-three and twelve base pairs; see figure 16.35.

35. A T-cell receptor is an immunoglobulinlike molecule located on the surfaces of T cells, enabling them to identify infected host cells.

37. One explanation is a defect in the maturation process of B cells. Another explanation is a defect in the process of V(D)J joining, which could involve five or more genes.

39. The simplest interpretation of these results is that something is different in the organization of antibody genes in embryonic cells and in B lymphocytes. If the genes were in the same place in both cases, we should have seen identical patterns for the two types of cells. The probe recognizes both variable and constant regions of the antibody gene, since it is made from the mature mRNA. In the B lymphocyte, the variable and constant regions are adjacent, but in the embryonic cells, there is some extra DNA between these two genes. This result led to the notion that variable genes are rearranged during the development of the immune system.

Critical Thinking Question:

1. Adenovirus attacks normal cells by binding to p53; it also attacks cancerous cells that lack p53. If we remove the gene for the protein that binds p53, the *E1B* gene, then the modified adenovirus will not be able to attack normal cells but will be able to attack cancerous ones lacking p53. This modification of adenovirus as a potential tool in treating cancer was published in 1996.

Chapter 17 Non-Mendelian Inheritance

1. Persistence of an environmentally induced trait into later generations is known as dauermodification and does not imply genetic control. After a suitable number of generations, the phenotype returns to normal, indicating an environmental rather than genetic response.
3. Maroon-like must affect the cytoplasm of the egg. The first cross is unusual and alerts us to maternal inheritance. A true Mendelian factor should produce one-half maroon-like males and females. The genotypes of the F_1 females are $ma\text{-}1^+/ma\text{-}1$ and $ma\text{-}1/ma\text{-}1$; half of the females should be of each genotype. If the wild-type allele produces wild-type cytoplasm for the progeny, regardless of the genotype of the progeny, any female that is $ma\text{-}1^+/ma\text{-}1$ will produce all wild-type progeny.
5. Although the genetic scheme predicts shell coiling perfectly, one could do experiments involving the injection of cytoplasm into eggs to test the viral hypothesis.
7. Female parent: *Dd.* Male parent: *d*–. (A): *dd.* Since selfing produces only sinistral snails, individual (A) must be homozygous for sinistral coiling, *dd.* The individual must get one *d* allele from each parent, so each parent must be at least heterozygous. Since (A) has dextral coiling, the mother must be heterozygous. The father could be either *D/d* or *d/d.*
9. By looking at different species, it is clear that few genes for oxidative phosphorylation are found in all mitochondrial genomes.
11. The rule of thumb is that suppressive petite mitochondria will dominate a cell, whereas neutral petite mitochondria will be lost in a competitive situation. Therefore

    segregational petite × segregational petite → segregational petites
    segregational petite × neutral petite → segregational petites
    segregational petite × suppressive petite → suppressive petites
    neutral petite × neutral petite → neutral petites
    neutral petite × suppressive petite → suppressive petites
    suppressive petite × suppressive petite → suppressive petites
13. In 0.02% of the offspring cells, the mt^- allele of the streptomycin locus is inherited. In essence, these cells seem to have inherited chloroplast genes from the mt^- parent. These cells thus provide us with a window on the possibility of having chloroplast genotypes from both parents viable within the same cell. Thus, interaction among other chloroplast genes can be looked for in this class of offspring (0.02% of total). If recombination occurs, map distances can be calculated by the usual methods, keeping in mind that we are taking data only from within this 0.02% of offspring.
15. Mitochondria and chloroplasts have prokaryotic affinities. They both have circular chromosomes, and their metabolism is affected by prokaryotic inhibitors (e.g., antibiotics). Certain prokaryotic messenger RNAs will hybridize with the organelle's DNA. There are numerous other aspects of biochemistry, morphology, and physiology that help to demonstrate affinities.
17. One way to determine that two loci are involved is to look at the proportion of offspring that lose mu particles after autogamy. In some strains, one-half the offspring will lose mu particles, indicating one locus was initially segregating (autogamy of $M_1m_1m_2m_2$ yields $M_1M_1m_2m_2$ or $m_1m_1m_2m_2$ in a 1:1 ratio). In other strains, one-fourth of the offspring will lose mu after autogamy, indicating that two unlinked loci were segregating (autogamy in $M_1m_1M_2m_2$ yields $M_1M_1M_2M_2$, $M_1M_1m_2m_2$, $m_1m_1M_2M_2$, or $m_1m_1m_2m_2$ in a 1:1:1:1 ratio).
19. Human mitochondrial DNA does not have introns. Finding an intron would suggest that the mitochondria had acquired a nuclear gene.
21. **a.** All type 1, 1.5 and 3.7 kilobases. **b.** All type 2, 2.5 and 6.0 kilobases. Recall that the chloroplast DNA from mt^- cells does not appear in progeny.
23. Conjugation produces exconjugants with the same genotype, but which haploid micronucleus survives in each cell is random. Therefore, one-fourth of the time, the genotypes of the exconjugants are expected to be *KK,* one-half of the time *Kk,* and one-fourth of the time *kk.* The sensitive exconjugant remains sensitive, regardless of which of the genotypes is present. Autogamy does not affect the genotype of homozygous exconjugants, but it does affect the heterozygote. Among the killer exconjugants we expect:

Killer Exconjugants	Autogamous Products	Phenotypic Ratio
1/4 *KK*	all *KK*	1/4 killer
1/2 *Kk*	1/2 *KK*	1/4 killer
	1/2 *kk*	1/4 sensitive (Kappa are lost)
1/4 *kk*	all *kk*	1/4 sensitive (Kappa are lost)

25. Use the striped plant as the egg parent, and get pollen from a plant with the following genotype: *IjIj jj.* If the striped plant is *iojap* (*ijij JJ*), all progeny will be heterozygous for both genes and will also contain *iojap* cytoplasm. The F_1 plants will segregate green, striped, and white sections within the plant. If the original plant is *IjIj jj* (and thus *japonica*), all F_1 plants will have striped leaves.
27. **a.** 2 normal:2 petite **b.** 0 normal:4 petite **c.** 4 normal: 0 petite. A nuclear gene should segregate 2:2 for each allele. Cytoplasmic factors will produce four spores with identical cytoplasm.

Critical Thinking Question:

1. We believe there are two mechanisms to ensure the distribution of cellular organelles during cytokinesis: stochastic and ordered inheritance. Stochastic inheritance simply means that no real mechanism exists; rather, the cell depends on the large number of the organelles to ensure an even distribution during the dividing of the cell. Ordered inheritance requires the even distribution of organelles in small numbers. This can be accomplished by special structures that divide a large organelle (e.g., a single, large chloroplast), or by other mechanisms that insert part of the mitochondrial system into new buds in budding yeast.

Chapter 18 Quantitative Inheritance

1. Three in two hundred is approximately 1 in $64 = 1/(4)^3$; therefore, three loci (see table 18.1). Each effective allele contributes about 1/2 pound over the 2-pound base (3-pound difference divided by six effective alleles: *AA BB CC* = 5 pounds, *aa bb cc* = 2 pounds).
3. Independent assortment: for example, *Aa Bb Cc Dd* parents can have *AA BB CC DD* offspring.
5. Individuals of intermediate color can produce both lighter and darker offspring by independent assortment. That is, *Aa Bb Cc Dd*

parents can produce *AA BB CC DD* and *aa bb cc dd* offspring. However, if each effective allele adds color, then individuals with the base color (white) who mate with each other (*aabbccdd*) presumably cannot have children with darker skin.

7. The marker stock is exposed to DDT over many generations, selecting for DDT resistance. It is then crossed with the wild-type and F_1 offspring are backcrossed to generate flies with various combinations of selected chromosomes. These flies are then tested for their DDT resistance. For example:

P_1 *Cy/Pm H/S Ce/M*(4) × + (wild-type)
F_1 (male) *Cy*/+ *H*/+ *Ce*/+ × (backcross) + (wild-type female)
F_2 *Cy*/+ +/+ +/+; +/+ H/+ +/+, etc.

or,

F_1 (male) *Cy*/+ *H*/+ *Ce*/+ × (backcross) *Cy/Pm H/S Ce/M*(*4*) (female)
F_2 *Cy/Pm H*/+ *Ce*/+; +/*Pm* +/*S* +/*M*(*4*); etc.

9. 50 cm. The difference between the two heights is 20 cm. This difference must result from the presence of effective (uppercase) alleles. The tall plant has four effective alleles, so each effective allele contributes an average of 5 cm to the height of the plant. The heterozygote has two effective alleles; $2 \times 5 = 10$ cm above the base of 40 cm, or a total of 50 cm.

11. Eight. The frequency of individuals in the F_2 that resemble one parent is $1/4^n$, where n = the number of genes involved. In this case, 100 of 6,200,000 were like one parent. 100/6,200,000 is approximately 1/64,000, which approximates $1/4^8$. So we probably have eight genes involved.

13. $r = 0.43$; $H_N = 0.43/0.50 = 0.86$ (narrow-sense heritability); environmental variance, a component of the total phenotypic variance appears in the denominator of the heritability equations (18.11 and 18.12). Thus, environmental factors that lower environmental variance (more uniform environments or environmental effects) increase heritability. And, factors that raise the environmental variance (less uniform environments or environmental effects) decrease heritability.

15. $H = (4 - 0.9)/(5 + 5) = 0.31$; H (high line) $= (4 - 3)/5 = 0.2$; H (low line) $= (3 - 0.9)/5 = 0.42$. Part of the difference may be due to the number of alleles available for selection in each direction and nonadditive factors.

17. First, "outstanding athletic ability" must be defined. It can be defined subjectively by accomplishment or more objectively with a physiological measure. Then genetic effects must be assessed through heritability analyses such as twin studies and correlations among relatives.

19. Uniformly good nutrition should increase the heritability of height by eliminating some of the environmental variance; it affects only the denominator in a heritability equation.

21. Thorax length: $V_D + V_I = (100 - [43 + 51]) = 6$; $H_N = 43/100 = 0.43$; $H_B = 49/100 = 0.49$. Eggs laid: $V_D + V_I = 44$; $H_N = 0.18$; $H_B = 0.62$.

23. 4.5 g.

$$H_N = \frac{\text{gain}}{\text{selection differential}}$$

Thus (H_N) (selection differential) = gain. Since $H_N = 0.5$ and selective differential = 9, (0.5)(9.0) = 4.5 g.

25. 176 pounds. To solve this problem, use the formula for realized heritability:

$$H = \frac{\text{gain}}{\text{selection differential}} = \frac{(Y_0 - \overline{Y})}{(Y_P - \overline{Y})}$$

$$0.4 = \frac{(Y_0 - \overline{Y})}{(Y_P - \overline{Y})} = \frac{Y_0 - \overline{Y}}{185 - 170}$$

$$Y_0 - \overline{Y} = (0.4)(15) = 6.0$$

$$6.0 = Y_0 - 170$$
$$Y_0 = 176 \text{ lbs}$$

Critical Thinking Question:

1. The simplest cause for retraction of a study is the failure of that study to be replicated by others. That would come about when the conclusion isn't generally supportable, a phenomenon that could have two major causes. First, the study might have been done well but the phenomenon was specific to that study, due possibly to small sample sizes or a "private mutation," an effect found in one family or a small group of individuals but not a general phenomenon. Second, there could have been an inadvertent error in the study. For example, a size difference in a small region of the brain of homosexual and heterosexual males was reported but has not been verified. One investigator, who is following up the study, suggested that the difference could be an effect of differences in the way the brains were preserved and thus represent no real (biological) effect.

Chapter 19 Population Genetics: The Hardy-Weinberg Equilibrium and Mating Systems

1. The frequencies of the three genotypes are $f(MM) = 41/100 = 0.41$; $f(MN) = 38/100 = 0.38$; $f(NN) = 21/100 = 0.21$. The frequency of *M*, *p*, is the frequency of *MM* homozygotes plus half the frequency of heterozygotes:

$$p = f(M) + (1/2)f(MN) = 0.41 + (1/2)(0.38) = 0.41 + 0.19 = 0.60$$
$$q = 1 - p = 1 - 0.60 = 0.40$$

Alternatively

$$p = f(M) = \frac{2(\#MM) + \#MN}{2 \times \text{total}} = \frac{2(41) + 38}{200}$$
$$= \frac{120}{200} = 0.60$$
$$q = 1 - p = 0.40$$

We do the following chi-square test:

	MM	*MN*	*NN*	Total
Observed	41	38	21	100
Expected	$p^2 \times 100$	$2pq \times 100$	$q^2 \times 100$	
	36	48	16	100
Chi-square	0.694	2.083	1.563	4.340

The critical chi-square (0.05, one degree of freedom) = 3.841. We thus reject the null hypothesis that this population is in Hardy-Weinberg proportions.

3. Here we must assume Hardy-Weinberg equilibrium because of dominance. The $f(tt) = 65/215 = 0.302$. Thus $q = f(t) = \sqrt{f(tt)} = \sqrt{0.302} = 0.55$; and $p = f(T) = 1 - 0.55 = 0.45$. Since there are zero degrees of freedom (number of phenotypes − number of alleles = 2 − 2 = 0), we cannot do a chi-square test to determine if the population is mating at random (in Hardy-Weinberg proportions).

5. a. Most human populations should be in Hardy-Weinberg proportions at the taster locus regardless of what the allelic frequencies are. 3:1 is a family ratio when the parents are heterozygotes or a population ratio when $p = q = 0.5$, given dominance.

b. With no other information, it is probably safest to assume $p = q = 0.5$. At equal frequencies of alleles, the dominant phenotype (tasting) occurs in 75% of people ($p^2 + 2pq$).

7. $p = 0.99, q = 0.01$. If the population is in equilibrium, there should be p^2 of AA + $2pq$ of Aa + q^2 of aa individuals. Since 1/10,000 shows the recessive trait, this is q^2. Therefore,

$$q = \sqrt{1/10{,}000} = \sqrt{0.0001} = 0.01$$

Since $p + q = 1, p = 1 - 0.01 = 0.99$.

9. The expected frequency of brown-eyed individuals will depend on the allelic frequencies of the original population. If we assume that mating is random with respect to eye color, and there is no selection, allelic frequencies will not change with time. We can, for example, calculate the frequencies of brown-eyed individuals for two populations at equilibrium. Let p = frequency of the brown-eye allele and q = frequency of the blue-eye allele.

Population	p	q	*Frequency of brown* ($p^2 + 2pq$)
1.	0.7	0.3	0.91
2.	0.5	0.5	0.75

We see that the original premise will be met only if the alleles are equally frequent.

11. 0.187 *MM,* 0.491 *MN,* 0.321 *NN.* The next generation will achieve equilibrium and there will be $(0.43)^2$ *MM* + $2(0.43)(0.57)$*MN* + $(0.57)^2$*MN.*

13. $f(A) = 0.792; f(B) = 0.208$. The easiest way to calculate frequencies is to do it empirically. We have three hundred people, so we have six hundred alleles.

$$f(A) = \frac{2 \times 200(AA) + 75(AB)}{600} = \frac{475}{600} = 0.792$$

$$f(B) = \frac{2 \times 25(BB) + 75(AB)}{600} = \frac{125}{600} = 0.208$$

15. a. Hardy-Weinberg proportions are achieved in one generation of random mating (multiple-allelic extension) if the locus is autosomal. If the locus is autosomal but there are different frequencies in the two sexes, then Hardy-Weinberg proportions are achieved in two generations. If the locus is sex linked, with different initial frequencies in the two sexes, then approach to equilibrium is gradual.

b. If the loci are not in equilibrium to begin with (linkage disequilibrium), then equilibrium is achieved asymptotically.

17. 0.63 type A, 0.08 type B, 0.28 type AB, 0.01 type O. Since the population is in equilibrium, the genotypic frequencies can be calculated as $(p + q + r)^2 = p^2 + 2pq + q^2 + 2pr + 2qr + r^2$. Let $p = 0.7$, $q = 0.2$, and $r = 0.1$. Blood types will be represented by the following:

A: $p^2 + 2pr$	B: $q^2 + 2qr$	AB: $2pq$	O: r^2
$= 0.49 + 0.14$	$= 0.04 + 0.04$	$= 0.28$	$= 0.01$

19. Inbreeding is disadvantageous when it causes recessive deleterious alleles to become homozygous. This occurs in normally outbred populations of diploids that have built up these harmful alleles. In species that normally inbreed, these deleterious alleles are probably no longer present; they were either removed by selection long ago or cannot build up in the population because of the regular pattern of inbreeding.

21. There are four paths passing through A, the only common ancestor ($F_A = 0.01$). All have ABCI as one side of the path. The second legs of the four other paths are ADEHI, ADGHI, AFGHI, and AFEHI. (A path such as IHEDAFGHI is invalid, passing through H twice.) Since each path has six ancestors, the inbreeding coefficient is $F_I = 4(1/2)^6(1.01) = 0.063$.

23. Using the formula $F = (2pq - H)/2pq$, we calculate that $2pq = 0.48$, and $H = 38/100 = 0.38$. Therefore, $F = (0.48 - 0.38)/0.48 = 0.208$.

25. 0.375. $F = (2pq - H)/2pq = \dfrac{(0.32 - 0.20)}{0.32} = \dfrac{0.12}{0.32} = 0.375$

Critical Thinking Question:

1. Let us use the superscripts m and p for male and female, respectively. Then, we can construct the following Punnett square creating the next generation:

	Males, $A; f(A) = p^m$	**Males,** $a; f(a) = q^m$
Females, $A; f(A) = p^f$	$f(AA) = p^m p^f$	$f(Aa) = p^f q^m$
Females, $a; f(a) = q^f$	$f(Aa) = p^m q^f$	$f(aa) = q^m q^f$

Since the distribution of offspring in the table is independent of sex, it is the same in both sexes. The frequency of the A allele, p (in both sexes), will be the sum of the frequencies of the homozygotes and half the heterozygotes, or:

$$p = p^m p^f + (1/2)(p^m q^f + p^f q^m)$$

We then substitute $(1 - p)$ for all q's:

$$p = p^m p^f + (1/2)p^m (1 - p^f) + (1/2)\, p^f (1 - p^m)$$

which simplifies to:

$$p = (1/2)(p^m + p^f)$$

In other words, after one generation of random mating, the allelic frequencies in each sex are the averages for both sexes. Now the frequency is the same in both sexes, and a second generation of random mating will achieve Hardy-Weinberg proportions. The population is not in equilibrium after one generation because the proportion of genotypes is not p^2, $2pq$, and q^2 if p^m does not equal p^f. In other words, $p^m p^f$ does not equal p^2.

Chapter 20 Population Genetics: Processes That Change Allelic Frequencies

1. a. The equilibrium frequency of a is $\hat{q} = \dfrac{\mu}{(\mu + \nu)}$. Therefore

$$\hat{q} = (6 \times 10^{-5})/(6 \times 10^{-5} + 7 \times 10^{-7}) = 0.00006/0.0000607 = 0.988$$

b. If $q = 0.90$ at generation n, then

$$q_{n+1} = q_n + \mu p_n - \nu q_n = 0.90 + (6 \times 10^{-5})(0.10) - (7 \times 10^{-7})(0.90) = 0.9000054$$

3. 0.714.

$$\hat{q} = \frac{\mu}{(\mu + \nu)} = \frac{5 \times 10^{-5}}{5 \times 10^{-5} + 2 \times 10^{-5}} = \frac{5}{7} = 0.714$$

5. We use equation 20.12 to calculate the migration rate, *m:*

$$m = (q_C - q_N)/(q_M - q_N)$$

In this case, $q_C = 0.45$, $q_N = 0.62$, and $q_M = 0.03$. Thus

$m = (0.45 - 0.62)/(0.03 - 0.62) = (-0.17)/(-0.59) = 0.288$

7. Fast: 0.56; slow: 0.44. With 900 butterflies we have 1,800 alleles. 0.6 (1,800) = 1,080 fast alleles, and 0.4 (1,800) = 720 slow alleles. In the migrant population, (0.8)(180) = 144 slow and (0.2)(180) = 36 fast alleles. Therefore, the frequency of the fast allele is (1,080 + 36)/1,980 = 0.56. The frequency of the slow allele is 1 − f(fast allele) = 1 − 0.56 = 0.44.

9. 0.47. We again use equation 20.12:

$$m = \frac{q_C - q_N}{q_M - q_N}$$

where $m = 0.1$, $q_C = 0.45$, and $q_M = 0.25$.

$$0.1 = \frac{0.45 - q_N}{0.25 - q_N}$$

$$0.1(0.25 - q_N) = 0.45 - q_N$$

$$0.025 - 0.1q_N = 0.45 - q_N$$

$$0.9q_N = 0.425$$

$$q_N = \frac{0.425}{0.9} = 0.47$$

11. In stabilizing selection, extremes of a distribution are selected against. In directional selection, one extreme is favored over the other. In disruptive selection, both extremes are favored over the middle of the distribution.

13. Heterozygote disadvantage:

	AA	*Aa*	*aa*	Total
Before selection	p^2	$2pq$	q^2	1
Fitnesses (*W*)	1	$1 - s$	1	
Frequencies after selection	$p^2/\overline{W}$	$2pq(1 - s)/\overline{W}$	$q^2/\overline{W}$	$\overline{W} = 1 - 2pqs$

Then

$$q_{n+1} = (pq[1 - s] + q^2)/\overline{W}$$

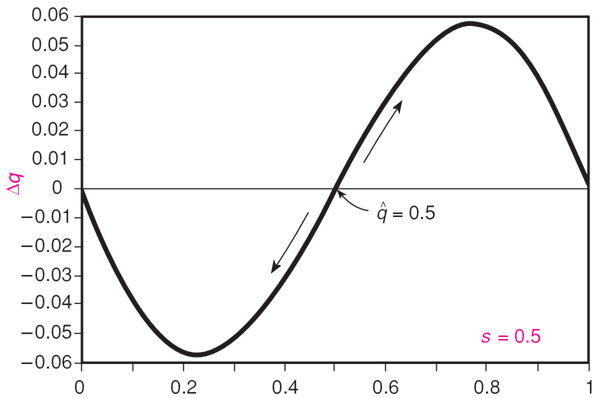
$$\Delta q = q_{n+1} - q = (pq[1 - s] + q^2)/\overline{W} - q\overline{W}/\overline{W}$$

$$= (pq[1 - s] + q^2 - q[1 - 2pqs])/\overline{W}$$

which simplifies to

$$\Delta q = spq(2q - 1)/\overline{W}$$

At $\Delta q = 0$, $\hat{q} = 0, 1$, or 0.5 ($2q - 1 = 0$, therefore $\hat{q} = 0.5$). The equilibrium points of zero and one are stable—if perturbed slightly (less than 0.5), the population will return to these values. The value $\hat{q} = 0.5$ is, however, unstable—if perturbed, it will continue away from the equilibrium point. This can be seen by either substituting into or graphing the Δq equation.



15.

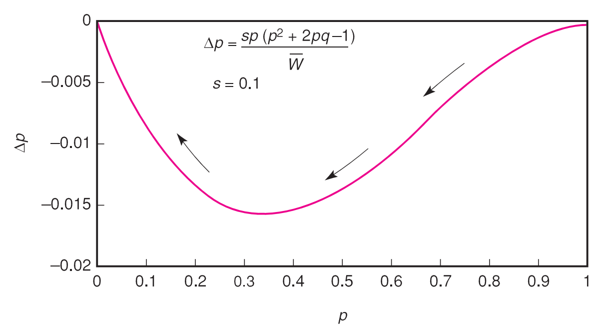
	AA	*Aa*	*aa*	Total
Before selection	p^2	$2pq$	q^2	1
Fitnesses (*W*)	$1 - s$	$1 - s$	1	$\overline{W} = 1 - p^2s - 2pqs$
Frequencies after selection	$p^2(1 - s)/\overline{W}$	$2pq(1 - s)/\overline{W}$	$q^2/\overline{W}$	1

$$p_{n+1} = (p^2[1 - s] + pq[1 - s])/\overline{W}$$

$$\Delta p = p_{n+1} - p = (p^2[1 - s] + pq[1 - s])/\overline{W} - p\overline{W}/\overline{W}$$

$$= sp(p^2 + 2pq - 1)/\overline{W} = sp(-p^2 + 2p - 1)/\overline{W}$$

And $\hat{p} = 0$ or 1 (from the root of the quadratic). Only zero is stable.



17. Since only heterozygotes survive, $\hat{q} = 0.5$. This can also be derived from equation 20.24:

$$\hat{q} = s_1/(s_1 + s_2)$$

If $s_1 = s_2 = 1$, then $\hat{q} = 1/2$

19. The *ST* inversion seems to do best at lower elevations and the *AR* at higher elevations. *CH* (and others) do not appear to be affected by altitude. To test this hypothesis, we would grow caged populations of flies with different initial frequencies of the various inversions at different simulated elevations, simulated by temperature, pressure, oxygen content, or other. We predict that, regardless of initial conditions, they would eventually equilibrate at the values in the table for the given parameter of the altitude (temperature, pressure, oxygen content, or other) that is acting as a selective agent. We would thus identify the selective agent. Since the inversions isolate various allelic combinations, our next step (a potentially long-term step) would be to determine which loci the selective agent is acting on.

21. 0.575. We can use the formula in equation 20.15, which is a simplified form of the sum of homozygotes + one-half the proportion of heterozygotes of the *b* allele.

$$q_{n+1} = \frac{q - sq^2}{1 - sq^2}$$

Since the relative fitness, $W = 0.4$, $s = 1 - W = 0.6$.

$$q_{n+1} = \frac{0.7 - (0.6)(0.49)}{1 - (0.6)(0.49)} = \frac{0.406}{0.706} = 0.575$$

23. 0.33, 0.25. Since the fitness is zero, $s = 1$, and we can use equation 20.17: $q_1 = q_0/(q_0 + 1) = 0.5/1.5 = 0.33$. For the second generation, we substitute the first generation numbers: $q_2 = 0.33/1.33 = 0.248$.

Critical Thinking Question:

1. There could be several reasons why these systems are in existence. First, they could be in selection-mutation equilibrium. However, that would not account for the high frequencies of both alleles in human populations in the Rh system. Second, the polymorphism could be relatively new, somehow maintaining both alleles as the human population increased in recent times with natural selection not having enough time to eliminate one of the alleles. Third, although the Rh blood system could follow the heterozygous disadvantage model, selection could also be acting in other ways that might maintain the polymorphism. That is, aside from Rh incompatibility eliminating heterozygotes, other genotypic combinations could be favored under other circumstances. Finally, although one or the other allele is being eliminated in any one population due to heterozygous disadvantage, the constant mixing of human populations could be reintroducing the rarer allele.

Chapter 21 Evolution and Speciation

1. Neo-Darwinism is the application of population genetics to Darwinian evolution. Darwinian evolution works as natural selection favors the most fit organisms in competition among the overproduced young of any species.

3. Each process lets reproductive isolating mechanisms evolve while some barrier to breeding arises (see fig. 21.3).

5. *Constraint* refers to the limitations on changes that can take place. Some changes result in nonfunctional proteins and enzymes and thus cannot be in a successful lineage. For example, many base changes that lead to new amino acids in enzyme active sites disrupt enzymatic activity. If these mutations take place, they are eliminated by natural selection.

7. Yes. Two distinct species should not yield fertile progeny. We see a great reduction in the numbers of offspring from hybrids, indicating that hybrid inviability is one isolating mechanism operating.

9. Punctuated equilibrium proposes that species remain unchanged for long periods of time and that major changes occur only periodically. If species A existed for ten million years and suddenly (geologically speaking) changed dramatically to species X and Y, there would be few fossils because of the relatively short time in which intermediate forms were present. Another argument is, barring an incredibly detailed and complete sequence (of which there are almost none), there will always be gaps in the fossil record.

11. Genetic variability can be maintained by heterozygote advantage (e.g., sickle-cell anemia in people); frequency-dependent selection (rare-male mating advantage in *Drosophila*); transient polymorphism (industrial melanism in moths during an increase or decrease in industrialization); life-stage selection, which often happens when comparing egg, larval, pupal, and adult mortalities in *Drosophila;* differential selection in heterogeneous environments, common in some land snails; and neutrality.

13. Presumably, in mammals, selection could involve types of substrates acted upon by electrophoretic variants; functioning at different pHs and ionic strengths in various cellular compartments; resistance to enzyme inhibitors; interaction with other proteins and membrane components; and others.

15. Use the formula $K = -\ln(1 - d/n)$, in which d is the number of amino acid differences and n is the total number of amino acid sites. Thus:

human being–dog: $K = -\ln(1 - 1/8) = 0.133$

human being–chicken: $K = -\ln(1 - 3/8) = 0.470$

and dog–chicken: $K = -\ln(1 - 3/8) = 0.470$

These values place human beings and dogs very close and both equally far from chickens, which is consistent with the known evolutionary relationships.

17. 0.33 *AA,* 0.49 *Aa,* 0.18 *aa.* The mean fitness of the population after selection is $0.5 + 2pq = 0.98$ (table 21.2). The new frequency of a genotype is its original frequency times its fitness, all divided by the mean fitness of the population. Or:

$$f(AA) = \frac{(0.36)(1.5 - 0.6)}{0.98} = 0.33$$

$$f(Aa) = 0.48/0.98 = 0.49$$

$$f(aa) = (0.16)(1.5 - 0.4)/0.98 = 0.18$$

19. 0.023. $K = -\ln(1 - p)$

$$K = -\ln(1 - [23/1{,}000]) = -\ln(0.0977) = 0.023$$

21. The third. For many amino acids, the third position can be any of the four bases. In addition, wobble allows for some variation here as well. Changing the first or second base almost always produces a new amino acid.

23. Eight first cousins, on average, carry the complete genome of an individual. Therefore, from an evolutionary point of view, an individual and his or her eight cousins include the same alleles.

Critical Thinking Question:

1. Given that the gene for peppering is recessive and that one year equals one generation, the moths should be following a selection model in which natural selection acts against the recessive homozygote. In that case, we have already developed a selection model for this in chapter 20. Equation 20.15 relates the new allelic frequency to the old, given a particular selection coefficient:

$$q_{n+1} = q(1 - sq)/(1 - sq^2)$$

We can solve this equation for the selection coefficient:

$$s = (q - q_{n+1})/(q^2 - q^2 q_{n+1})$$

Here, $q = 0.6$ and $q_{n+1} = 0.5$. When we solve the equation, we get: $s = 0.56$, or $W = 1 - s = 0.44$, which is the fitness of the peppered moth.

APPENDIX B

Suggestions for Further Reading

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Chapter 14 Gene Expression: Control in Prokaryotes and Phages

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Chapter 15 The Eukaryotic Chromosome

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Chapter 16 Gene Expression: Control in Eukaryotes

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GLOSSARY

acentric fragment A chromosomal piece without a centromere.
acrocentric chromosome A chromosome whose centromere lies very near one end.
acron The anterior end of the arthropod embryo from which eyes and antennae develop.
activation energy *See* free energy of activation ($\Delta G^{\ddagger}$).
activator A eukaryotic specific transcription factor that binds to an enhancer, often far upstream of a promoter.
active site The part of an enzyme where the actual enzymatic function is performed.
adaptive mutation *See* directed mutation.
adaptive value *See* fitness.
additive model A mechanism of quantitative inheritance in which alleles at different loci either add an amount to the phenotype or add nothing.
adenine *See* purines.
adjacent-1 segregation A separation of centromeres during meiosis in a reciprocal translocation heterozygote so that unbalanced zygotes are produced.
adjacent-2 segregation Separation of centromeres during meiosis in a translocation heterozygote so that homologous centromeres are pulled to the same pole.
A DNA The form of DNA with high water content; it has tilted base pairs and more base pairs per turn than does B DNA.
affected Individuals in a pedigree that exhibit the specific phenotype under study.
allele Alternative form of a gene.
allelic exclusion A process whereby only one immunoglobulin light chain and one heavy chain gene are transcribed in any one cell; the other genes are repressed.
allopatric speciation Speciation in which the evolution of reproductive isolating mechanisms occurs during the physical separation of the populations.
allopolyploidy Polyploidy produced by the hybridization of two species.
allosteric protein A protein whose shape is changed when it binds a particular molecule. In the new shape, the protein's ability to react to a second molecule is altered.
allotype Mutant of a nonvariant part of an immunoglobulin gene that follows the rules of simple Mendelian inheritance.
allozygosity Homozygosity in which the two alleles are alike but unrelated. *See* autozygosity.
allozymes Forms of an enzyme, controlled by alleles of the same locus, that differ in electrophoretic mobility. *See* isozymes.
alternate segregation A separation of centromeres during meiosis in a reciprocal translocation heterozygote so that balanced gametes are produced.
alternative splicing Various ways of splicing out introns in eukaryotic premessenger RNAs so that one gene produces several different messenger RNA and protein products.
altruism A form of behavior in which an individual risks lowering its fitness for the benefit of another.
Alu family A dispersed, intermediately repetitive DNA sequence found in the human genome about 300,000 times. The sequence is about 300 bp long. The name *Alu* comes from the restriction endonuclease that cleaves it.
aminoacyl-tRNA synthetases Enzymes that attach amino acids to their proper transfer RNAs.
amphidiploid An organism produced by hybridization of two species, followed by somatic doubling. It is an allotetraploid that appears to be a normal diploid.
anagenesis The evolutionary process whereby one species evolves into another without any splitting of the phylogenetic tree. *See* cladogenesis.
anaphase The stage of mitosis and meiosis in which sister chromatids or homologous chromosomes are separated by spindle fibers.
anaphase A The stage of anaphase in which chromatids are separated by the shortening of kinetochore microtubules.
anaphase B The stage of anaphase in which chromatids are separated by the general elongation of the spindle.
anaphase-promoting complex (APC) Protein complex that breaks down cyclin B and promotes anaphase among its various roles in controlling the cell cycle. (Also called the cyclosome.)
aneuploids Individuals or cells exhibiting aneuploidy.
aneuploidy The condition of a cell or of an organism that has additions or deletions of whole chromosomes.
angiosperms Plants whose seeds are enclosed within an ovary. Flowering plants.
antibody A protein produced by a B lymphocyte that protects the organism against antigens.
anticoding strand The DNA strand that forms the template for both the transcribed messenger RNA and the coding strand.
anticodon The three-base sequence on transfer RNA complementary to a codon on messenger RNA.
antigen A foreign substance capable of triggering an immune response in an organism.
antimutator mutations Mutations of DNA polymerase that decrease the overall mutation rate of a cell or of an organism.
anti-oncogene A gene that represses malignant growth and whose absence results in malignancy (e.g., *retinoblastoma*).
antiparallel strands Strands, as in DNA, that run in opposite directions with respect to their 3′ and 5′ends.
antisense RNA RNA product of *mic* (*m*RNA-*i*nterfering *c*omplementary RNA) genes that regulates another gene by base pairing with, and thus blocking, its messenger RNA.
antisense strand *See* anticoding strand.
anti-sigma factor A protein that interferes with the action of a sigma factor.
antiterminator protein A protein that, when bound at its normal attachment sites, lets RNA polymerase read through normal terminator sequences (e.g., the N- and Q-gene products of phage λ).
AP endonucleases Endonucleases that initiate excision repair at apurinic and apyrimidinic sites on DNA.
apoptosis Programmed cell death.
archaea Highly specialized, bacterialike organisms that make up the third kingdom of life on earth along with the bacteria and the eukaryotes. Identified by Carl Woese in 1977 based on ribosomal RNA sequences. Most are thermophilic, halophilic, or methanogenic.
ascospores Haploid spores found in the asci of Ascomycete fungi.
ascus The sac in Ascomycete fungi that holds the ascospores.
A (aminoacyl) site The site on the ribosome occupied by an aminoacyl-tRNA just prior to peptide bond formation.
assignment test A test that determines whether a locus is on a specific chromosome by observing the concordance of the locus and the specific chromosome in hybrid cell lines.
assortative mating The mating of individuals with similar phenotypes.

aster Configuration at the centrosome with microtubules radiating out in all directions.
ataxia-telangiectasia A disease in human beings caused by a defect in X-ray-induced repair mechanisms.
attenuator region A control region at the promoter end of repressible amino acid operons that exerts transcriptional control based on the translation of a small leader peptide gene.
attenuator stem *See* terminator stem.
autogamy Nuclear reorganization in a single *Paramecium* cell similar to the changes that occur during conjugation.
autonomously replicating sequence (ARS) Eukaryotic site of the initiation of DNA replication consisting of an 11 base-pair consensus sequence and several other sequences covering 100–200 base pairs.
autopolyploidy Polyploidy in which all the chromosomes come from the same species.
autoradiography A technique in which radioactive molecules make their locations known by exposing photographic plates.
autosomal set A combination of nonsex chromosomes consisting of one from each homologous pair in a diploid species.
autosomes The nonsex chromosomes.
autotrophs Organisms that can utilize carbon dioxide as a carbon source.
autozygosity Homozygosity in which the two alleles are identical by descent (i.e., they are copies of an ancestral gene).
auxotrophs Organisms that have specific nutritional requirements.
bacillus A rod-shaped bacterium.
backcross The cross of an individual with one of its parents or with an organism with the same genotype as a parent.
back mutation The process that causes reversion. A change in a nucleotide pair in a mutant gene that restores the original sequence and hence the original phenotype.
bacterial artificial chromosomes (BACs) Artificial chromosomes used for sequencing that are derived from bacterial fertility factors (F plasmids).
bacterial lawn A continuous cover of bacteria on the surface of a growth medium.
bacteriophages Bacterial viruses.
Balbiani rings The larger polytene chromosomal puffs. Generally synonymous with *puffs*. *See* chromosome puffs.
Barr body Heterochromatic body (X chromosome) found in the nuclei of normal female mammals but absent in the nuclei of normal males.
basal bodies Microtubule organizing center for cilia and flagella, composed of centrioles.
base excision repair The DNA excision repair mechanism that replaces a nucleotide lacking its base with a complete nucleotide. Glycosylases or the environment create the AP (apurinic and apyrimidinic) nucleotides.
base flipping A process whereby enzymes gain access to bases within the DNA double helix by first flipping the bases out of the interior to the outside.
basic/helix-loop-helix/leucine zipper A motif of proteins that bind DNA that consists of a series of basic amino acids followed by a helix-loop-helix domain and then a leucine zipper. *See* helix-turn-helix motif; leucine zipper.
Batesian mimicry Form of mimicry in which an innocuous model gains protection by resembling a noxious or dangerous host.
B DNA The right-handed, double-helical form of DNA described by Watson and Crick.
β-galactosidase The enzyme that splits lactose into glucose and galactose (coded by a gene in the *lac* operon).
β-galactoside acetyltransferase An enzyme that is involved in lactose metabolism and encoded by a gene in the *lac* operon.
β-galactoside permease An enzyme involved in concentrating lactose in the cell (coded by a gene in the *1ac* operon).
binary fission Simple cell division in single-celled organisms.
binomial expansion The terms generated when a binomial expression is raised to a particular power.
binomial theorem The theorem that gives the terms of the expansion of a binomial expression raised to a particular power.
biochemical genetics The study of the relationships between genes and enzymes, specifically the role of genes in controlling the steps in biochemical pathways.
bioinformatics The science of storing, retrieving, and analyzing genomic data.
biolistic A method (*bio*logical bal*listic*) of transfecting cells by bombarding them with microprojectiles coated with DNA.
biological species concept The idea that organisms are classified in the same species if they are potentially capable of interbreeding and producing fertile offspring.
bivalents Structures, formed during prophase of meiosis I, consisting of the synapsed homologous chromosomes. Equivalent to a tetrad of chromatids.
blastoderm The outer layer of cells in an insect embryo after cleavage but before gastrulation. A syncitial blastoderm gives way to a cellular blastoderm when cell walls form.
blunt-end ligation The ligating or attaching of blunt-ended pieces of DNA by, for example, T4 DNA ligase. Used in creating hybrid vectors.
bottleneck A brief reduction in the size of a population, which usually leads to random genetic drift.
bouquet stage A stage during zygonema in which chromosome ends, attached to the nuclear membrane, come to lie near each other.
branch migration The process in which a crossover point between two duplexes slides along the duplexes.
breakage and reunion The general mode by which recombination occurs. DNA duplexes are broken and reunited in a crosswise fashion according to the Holliday model.
breakage-fusion-bridge cycle Damage that a dicentric chromosome goes through during each cell cycle.
buoyant density of DNA A measure of the density or size of DNA determined by the equilibrium point reached by DNA after density gradient centrifugation.
calculus of the genes Apparent calculation by the genes to determine when a particular altruistic behavior is beneficial to inclusive fitness and hence worth doing.
cancer An informal term for a diverse class of diseases marked by abnormal cell proliferation.
cancer-family syndromes Pedigree patterns in which unusually large numbers of blood relatives develop certain kinds of cancers.
cap A methylated guanosine added to the 5′ end of eukaryotic messenger RNA.
capsid The protein shell of a virus.
capsomere Protein clusters making up discrete subunits of a viral protein shell.
carcinoma Tumor arising from epithelial tissue (e.g., glands, breasts, skin, linings of the urogenital and respiratory systems).
cassette mechanism The mechanism by which homothallic yeast cells alternate mating types. The mechanism involves two silent transposons (cassettes) and a region where these cassettes can be expressed (cassette player).
catabolite activator protein (CAP) A protein that, when bound with cyclic AMP, can attach to sites on sugar-metabolizing operons to enhance transcription of these operons.
catabolite repression Repression of certain sugar-metabolizing operons in favor of glucose utilization when glucose is present in the environment of the cell.
cDNA *See* complementary DNA.
cell cycle The cycle of cell growth, replication of the genetic material, and nuclear and cytoplasmic division.
cell-free system A mixture of cytoplasmic components from cells, lacking nucleic acids and membranes. Used for in vitro protein synthesis and other purposes.
cellular immunity Immunity controlled by killer and helper T cells, which recognize infected body cells and either cause the infected cells to destroy internal invaders or destroy the infected cells directly.
centimorgan A chromosome-mapping unit. One centimorgan equals 1% recombinant offspring.

central dogma The original postulate of the way that information can be transferred between DNA, RNA, and protein, barring any transfer originating in protein.
centric fragment A piece of chromosome containing a centromere.
centrioles Cylindrical organelles, found in eukaryotes (except in higher plants), that reside in the centrosome. Also called basal bodies when they organize flagella or cilia.
centromere Constrictions in eukaryotic chromosomes on which the kinetochore lies. Also, the DNA sequence within the constriction that is responsible for appropriate function.
centromeric fission Creation of two chromosomes from one by splitting at the centromere.
centrosome The spindle-microtubule organizing center in eukaryotes except for those, such as fungi, that use spindle pole bodies to organize the spindles.
chaperone *See* molecular chaperone.
Chargaff's rule Chargaff's observation that in the base composition of DNA, the quantity of adenine equals the quantity of thymine, and the quantity of guanine equals the quantity of cytosine (equal purine and pyrimidine content).
Charon phages Phage lambda derivatives used as vehicles in DNA cloning.
checkpoint Used to describe points in the cell cycle that can be stopped if certain conditions are not met.
chemiluminescent techniques Techniques in which various molecules are made visible when exposed to ultraviolet or laser light if the molecules have a fluorescent tag.
chiasmata X-shaped configurations seen in tetrads during the later stages of prophase I of meiosis. They represent physical crossovers (singular: *chiasma*).
chimeras Individuals made up of two or more cell lines in which the cells originated in different zygotes. *See* mosaics.
chimeric plasmid Hybrid, or genetically mixed, plasmid used in DNA cloning.
chi site Sequence of DNA at which the RecBCD protein cleaves one of the DNA strands during recombination.
chi-square distribution The sampling distribution of the chi-square statistic. A family of curves whose shapes depend on degrees of freedom.
chloroplast The organelle that carries out photosynthesis and starch grain formation.
chromatids The subunits of a chromosome prior to anaphase of meiosis or mitosis. At anaphase of meiosis II or mitosis, when the sister chromatids separate, each chromatid becomes a chromosome.
chromatin The nucleoprotein material of the eukaryotic chromosome.
chromatin assembly factors Proteins involved in the construction of nucleosomes.
chromatin remodeling The change in the structure or positioning of nucleosomes, usually to allow transcription.
chromatosome The core nucleosome plus the Hl protein, a unit that includes approximately 168 base pairs of DNA.
chromomeres Dark regions of chromatin condensation in eukaryotic chromosomes at meiosis, mitosis, or endomitosis.
chromosomal painting A variant of the technique known as fluorescent in situ hybridization. Fluorescent dyes, attached to numerous nucleotide probes, give each human chromosome a different fluorescent signature.
chromosomal theory of inheritance The theory that chromosomes are linear sequences of genes.
chromosome The form of the genetic material in viruses and cells. A circle of DNA in most prokaryotes; a DNA or an RNA molecule in viruses; a linear nucleoprotein complex in eukaryotes.
chromosome jumping A technique for isolating clones from a genomic library that are not contiguous but skip a region between known points on the chromosome. This is usually done to bypass regions that are difficult or impossible to "walk" through or regions known not to be of interest.
chromosome puffs Diffuse, uncoiled regions in polytene chromosomes where transcription is actively taking place.
chromosome walking A technique for studying segments of DNA, larger than can be individually cloned, by using overlapping cloned DNA.
cis Meaning "on the near side of"; refers to geometric configurations of atoms or mutants on the same chromosome.
***cis*-dominant** Mutants (e.g., of an operator) that control the functioning of genes on the same piece of DNA.
***cis-trans* complementation test** A mating test to determine whether two mutants on opposite chromosomes complement each other; a test for allelism.
cistron Term Benzer coined for the smallest genetic unit that exhibits the *cis-trans* position effect; synonymous with *gene*.
cladogenesis The evolutionary process whereby one species splits into two or more species. *See* anagenesis.
classical linkage map A chromosomal map, measured in centimorgans, based on genetic crosses to locate the relative distances between genes and their relative locations on chromosomes.
clinal selection Selection that changes gradually along a geographic gradient.
clonal evolution theory The theory that cancer develops from sequential changes (mutations) in the genome of a single cell.
clone A group of cells arising from a single ancestor.
coccus A spherical bacterium.
coding strand The DNA strand with the same sequence as the transcribed messenger RNA (given U in RNA and T in DNA). Compare with "anticoding strand."
codominance The relationship of alleles in a heterozygote that shows the individual expression of each allele in the phenotype.
codon preference The idea that for amino acids with several codons, one or a few are preferred and are used disproportionately. They would correspond with abundant transfer RNAs.
codons The sequences of three RNA or DNA nucleotides that specify either an amino acid or termination of translation.
coefficient of coincidence The number of observed double crossovers, divided by the number expected based on the independent occurrence of crossovers.
coefficient of relationship, *r* The proportion of alleles held in common by two related individuals.
cohesin Proteinaceous complex that holds sister chromatids together until anaphase of mitosis.
cointegrate A fusion of two elements. An intermediate structure in transposition.
colicinogenic factors *See* col plasmids.
col plasmids Plasmids that produce antibiotics (colicinogens) that the host uses to kill other strains of bacteria.
combinatorial control Transcriptional control in eukaryotes, which involves a large number of polypeptides, many of which recognize specific DNA sequences.
common ancestry The shared genetic inheritance of two individuals who are blood relatives. When two parents have a common ancestor, their offspring will be inbred.
compensasome The multisubunit dosage compensation complex in *Drosophila* with attendant RNAs.
competence factor A surface protein that binds extracellular DNA and enables the bacterial cell to be transformed.
complementarity The correspondence of DNA bases in the double helix so that adenine in one strand is opposite thymine in the other strand and cytosine in one strand is opposite guanine in the other. This relationship explains Chargaff's rule.
complementary DNA (cDNA) DNA synthesized by reverse transcriptase using RNA as a template.
complementation The production of the wild-type phenotype by a cell or an organism that contains two mutant genes. If complementation occurs, the mutants are almost certainly nonallelic.
complementation group Cistron (determined by the *cis-trans* complementation test).
complete medium A culture medium that is enriched to contain all of the growth requirements of a strain of organisms.

component of fitness A particular aspect in the life cycle of an organism upon which natural selection acts.
composite transposon A transposon constructed of two IS elements flanking a control region that frequently contains host genes.
concordance The amount of phenotypic similarity between individuals.
condensin A protein complex including SMC (structural maintenance of chromosomes) proteins needed for the condensation of interphase chromosomes to mitotic chromosomes.
conditional-lethal mutant A mutant that is lethal under one condition but not lethal under another.
confidence limits A statistical term for a pair of numbers that predicts the range of values within which a particular parameter lies.
conjugation A process whereby two cells come in contact and exchange genetic material. In prokaryotes, the transfer is a one-way process.
consanguineous Meaning "between blood relatives"; usually refers to inbreeding or incestuous matings.
consensus sequence A sequence of the common nucleotides found in many different DNA or RNA samples of homologous regions (e.g., promoters).
conservative replication A postulated mode of DNA replication in which an intact double helix acts as a template for a new double helix; known to be incorrect.
conserved sequence A sequence found in many different DNA or RNA samples (e.g., promoters) that is invariant in the sample.
constitutive heterochromatin Heterochromatin that surrounds the centromere. *See* satellite DNA.
constitutive mutant A mutant whose transcription is no longer under regulatory control.
contigs Genomic libraries of overlapping, contiguous clones that cover complete regions of a chromosome.
continuous replication In DNA, uninterrupted replication in the 5′ to 3′ direction using a 3′ to 5′ template.
continuous variation Variation measured on a continuum rather than in discrete units or categories (e.g., height in human beings).
corepressor The metabolite that when bound to the repressor (of a repressible operon) forms a functional unit that can bind to its operator and block transcription.
correlation coefficient A statistic that gives a measure of how closely two variables are related.
cosmid A hybrid plasmid that contains *cos* sites at each end. *Cos* sites are recognized during head filling of lambda phages. Cosmids are useful for cloning segments of foreign DNA up to 50 kb.
cotransduction The simultaneous transduction of two or more genes.
coupling Allelic arrangement in which mutants are on the same chromosome and wild-type alleles are on the homologue.
covariance A statistical value measuring the simultaneous deviations of *x* and *y* variables from their means.
CpG islands Stretches of CG repeats (in which CpG indicates sequential bases on the same strand of DNA, rather than a C-G base pair). These repeats, found in imprinting centers, are important in regulation.
crisscross pattern of inheritance The phenotypic pattern of inheritance controlled by X-linked recessive alleles.
critical chi-square A chi-square value for a given degree of freedom and probability level, to which we can compare an experimental chi-square.
crossbreed To facilitate fertilization between separate individuals.
cross-fertilization *See* crossbreed.
crossing over A process in which homologous chromosomes exchange parts by a breakage-and-reunion process.
crossovers *See* chiasmata.
crossover suppression The apparent lack of crossing over within an inversion loop in heterozygotes. Usually due to mortality of zygotes carrying defective crossover chromosomes rather than to actual suppression.
C-value paradox Structural and junk DNA create large eukaryotic genomes and large differences in DNA content between eukaryotic species.
cyclic AMP A form of AMP (adenosine monophosphate) used frequently as a second messenger in eukaryotic hormone nets and in catabolite repression in prokaryotes.
cyclin Family of proteins involved in cell cycle control.
cyclin-dependent kinase (CDK) Family of kinases (phosphorylating enzymes) that, when combined with cyclin, are active in controlling checkpoints in the cell cycle.
cyclosome *See* anaphase-promoting complex (APC).
cytogenetics The study of cells from the perspective of genetics. In practice, the study of changes in the gross structure and number of chromosomes in cells.
cytokinesis The division of the cytoplasm of a cell into two daughter cells. *See* karyokinesis.
cytoplasmic inheritance Extrachromosomal inheritance controlled by nonnuclear genomes.
cytosine *See* pyrimidines.
cytotoxic T lymphocytes T cells responsible for attacking host cells that have been infected with an invading bacterium or virus.
dauermodification The persistence for several generations of an environmentally induced trait.
degenerate code A code in which several code words have the same meaning. The genetic code is degenerate because many different codons may specify the same amino acid.
degrees of freedom An estimate of the number of independent categories in a particular statistical test or experiment.
deletion chromosome A chromosome with part deleted.
deme A locally interbreeding population.
denatured Loss of natural configuration (of a molecule) through heat or other treatment. Denatured DNA is single-stranded.
denominator elements Genes on the autosomes of *Drosophila* that regulate the sex switch (*sxl*) to the off condition (maleness). Refers to the denominator of the X/A genic balance equation.
density-gradient centrifugation A method of separating molecular entities by their differential sedimentation in a centrifugal gradient.
depauperate fauna A fauna, especially common on islands, lacking many species found in similar habitats.
derepressed The condition of an operon that is transcribing because repressor control has been lifted.
deterministic Referring to events that have no random or probabilistic aspects but proceed in a fixed, predictable fashion.
development The process of orderly change an individual goes through in the formation of structure.
diakinesis The final stage of prophase I of meiosis.
dicentric chromosome A chromosome with two centromeres.
dictyotene A prolonged diplonema of primary oocytes that can last many years.
dideoxy method A method of DNA sequencing that uses chain-terminating (dideoxy) nucleotides.
dihybrid An organism heterozygous at two loci.
dimerization The chemical union of two similar molecules.
diploid Having each chromosome in two copies per nucleus or cell.
diplonema (diplotene stage) The stage of prophase of meiosis I in which chromatids appear to repel each other.
directed mutation A form of mutation that appears to respond to the needs of the cell but may, in fact, be due to the cell's hypermutable state under duress.
directional selection A type of selection that removes individuals from one end of a phenotypic distribution and thus causes a shift in the distribution.
disassortative mating The mating of two individuals with dissimilar phenotypes.

discontinuous replication In DNA, the replication in short 5′ to 3′ segments, using the 5′ to 3′ strand as a template while going backward, away from the replication fork.

discontinuous variation Variation that falls into discrete categories (e.g., the green and yellow color of garden peas).

discrete generations Generations that have no overlapping reproduction. All reproduction takes place between individuals of the same cohort.

dispersive replication A postulated mode of DNA replication combining aspects of conservative and semiconservative replication; known to be incorrect.

disruptive selection A type of selection that removes individuals from the center of a phenotypic distribution and thus causes the distribution to become bimodal.

D-loop Configuration found during DNA replication of chloroplast and mitochondrial chromosomes wherein the origin of replication is different on the two strands. The first structure formed is a displacement loop, or D-loop.

DNA cloning *See* gene cloning.

DNA-DNA hybridization The process of taking DNA from the same or different sources and heating and then cooling it, causing double helices to re-form at homologous regions. This technique is useful for determining sequence similarities and degrees of repetitiveness among DNAs.

DNA fingerprint A pattern of bands created on an electrophoretic gel of a DNA digest probed for a variable locus.

DNA glycosylases Endonucleases that initiate excision repair at the sites of various damaged or improper bases in DNA.

DNA gyrase A topoisomerase that relieves supercoiling in DNA by creating a transient break in the double helix.

DNA ligase An enzyme that closes nicks or discontinuities in one strand of double-stranded DNA by creating an ester bond between adjacent 3′-OH and 5′-PO_4 ends on the same strand.

DNA polymerase One of several classes of enzymes that polymerize DNA nucleotides using single-stranded DNA as a template.

DNA-RNA hybridization The process of heating and then cooling a mixture of DNA and RNA so that the RNA can hybridize (form a double helix) with DNA with a complementary nucleotide sequence.

docking protein Responsible for attaching (docking) a ribosome to a membrane by interacting with a signal particle attached to a ribosome destined to be membrane bound.

dominant An allele that expresses itself even when heterozygous. Also, the trait controlled by that allele.

dosage compensation A mechanism by which species with sex chromosomes ensure that one sex does not have differential activity of alleles on the sex chromosomes.

dot blotting A blotting technique, used on DNA already cloned, that eliminates the electrophoretic separation step. Autoradiographs reveal dots rather than bands on a gel, indicating a probed sequence.

double digest The product formed when two different restriction endonucleases act on the same segment of DNA.

double helix The normal structural configuration of DNA consisting of two helices rotating about the same axis.

downstream A convention on DNA related to the position and direction of transcription by RNA polymerase (5′→3′). Downstream (in the 3′ direction) is in the direction of transcription, whereas upstream (in the 5′ direction) is in the direction from which the polymerase has come.

downstream promoter element (DPE) A consensus sequence at about +28 to +34 of RNA polymerase II promoters that have initiator elements but not TATA boxes.

dyad Two sister chromatids attached to the same centromere.

dynein Microtubule motor protein.

dysplasia Excessive cell growth that involves pathological changes to the cells and their nuclei.

electrophoresis The separation of molecular entities by electric current.

electroporation A technique for transfecting cells by applying a high-voltage electric pulse.

elongation complex The form of RNA polymerase II that actively carries out basal transcription.

elongation factors (EF-Ts, EF-Tu, EF-G) Proteins necessary for the proper elongation and translocation processes during translation at the ribosome in prokaryotes. Replaced by eEFIα and eEFIβγ in eukaryotes.

endogenote Bacterial host chromosome.

endomitosis Chromosomal replication without nuclear or cellular division that results in cells with many copies of the same chromosome, such as in the salivary glands of *Drosophila*.

endonucleases Enzymes that make nicks internally in the backbone of a polynucleotide. They hydrolyze internal phosphodiester bonds.

enhancer A eukaryotic DNA sequence that increases transcription of a gene by binding specific transcription factors.

enriched medium *See* complete medium.

enzyme Protein catalyst.

epigenetic effect An environmentally induced change in the genetic material that does not cause a change in base pairs. Generally, a phenomenon of differential expression of alleles of a locus depending on the parent of origin. Also applied to an effect in proteins.

epistasis The masking of the action of alleles of one gene by allelic combinations of another gene.

equational division A division, such as the second meiotic division, that does not reduce chromosomal numbers.

E (exit) site Site on the ribosome that depleted transfer RNAs pass through during ejection.

euchromatin Regions of eukaryotic chromosomes that are diffuse during interphase. Presumably the actively transcribing DNA of the chromosomes.

eugenics A social movement designed to improve humanity by encouraging those with beneficial traits to breed and discouraging those with undesirable traits from breeding.

eukaryotes Organisms with true nuclei.

euploidy The condition of a cell or organism that has one or more complete sets of chromosomes.

evolution A change in phenotypic frequencies in a population.

evolutionary rates The rate of divergence between taxonomic groups, measurable as number of amino acid substitutions per million years.

excision repair A process whereby cells remove part of a damaged DNA strand and replace it through DNA synthesis, using the undamaged strand as a template.

exconjugant Each of the two cells that separate after conjugation has taken place.

exogenote DNA that a bacterial cell has taken up through one of its sexual processes.

exon In a gene that has intervening sequences (introns), a region that is actually exported from the nucleus to be expressed or become part of a transfer or ribosomal RNA.

exon shuffling The hypothesis put forward by Walter Gilbert that exons code for the functional units of a protein, and that the evolution of new genes proceeds by recombination or the exclusion of exons.

exonucleases Enzymes that digest nucleotides from the ends of polynucleotide molecules. They hydrolyze phosphodiester bonds of terminal nucleotides.

experimental design A branch of statistics that attempts to outline the way in which experiments should be carried out so the data gathered has statistical value.

expression vector A hybrid vector (plasmid) that expresses its cloned genes.

expressivity The degree of expression of a genetically controlled trait.

F_1 *See* filial generation.

factorial The product of all integers from the specified number down to one (unity).

Fanconi's anemia A disease in human beings with a syndrome of congenital malformations; associated with various cancers.

fate map A map of the developmental fate of a zygote or early embryo showing the adult organs that will develop from a given position on the zygote or early embryo.

F-duction *See* sexduction.
fecundity selection The forces causing one genotype to be more fertile than another genotype.
feedback inhibition A posttranslational control mechanism in which the end product of a biochemical pathway inhibits the activity of the first enzyme of the same pathway.
fertility factor The plasmid that allows a prokaryote to engage in conjugation with, and pass DNA into, an F^- cell.
F factor *See* fertility factor.
filial generation Offspring generation. F_1 is the first offspring, or filial, generation; F_2 is the second; and so on.
fimbriae *See* pili.
first-division segregation (FDS) The allelic arrangement of ordered spores that indicates the lack of recombination between a locus and its centromere.
fitness, *W* The relative reproductive success of a genotype as measured by survival, fecundity, or other life-history parameters.
5′ untranslated region (5′ UTR) *See* leader.
floral meristem The shoot apical meristem sets aside this tissue that gives rise to flowers.
floral-meristem identity genes At least five genes known to establish the identity of the floral meristem.
fluctuation test An experiment by Luria and Delbrück that compared the variance in number of mutations among small cultures with that among subsamples of a large culture to determine the mechanism of inherited change in bacteria.
fluorescent in situ hybridization (FISH) A technique in which a fluorescent dye is attached to a nucleotide probe that then binds to a specific site on a chromosome and makes itself visible by its fluorescence.
Fokker-Planck equation An equation that describes diffusion processes. It is used by population geneticists to describe random genetic drift.
footprinting A technique to determine the length of nucleic acid in contact with a protein. While in contact, the free DNA is digested. The remaining DNA is then isolated and characterized.
founder effect Genetic drift observed in a population founded by a small, nonrepresentative sample of a larger population.
F-pili Sex pili. Hairlike projections on an F^+ or Hfr bacterium involved in anchorage during conjugation.
fragile site A chromosomal region that has a tendency to break.
fragile-X syndrome The most common form of inherited mental retardation. Named for its association with an X chromosome with a tip that breaks or appears uncondensed. Inheritance involves imprinting.
frameshift A mutation in which there is an addition or deletion of nucleotides that causes the codon reading frame to shift.
free energy of activation ($\Delta G^{\ddagger}$) Energy needed to initiate a chemical reaction.
frequency-dependent selection A selection whereby a genotype is at an advantage when rare and at a disadvantage when common.
functional alleles Mutations that fail to complement each other in a *cis-trans* complementation test.
fundamental number (NF) The number of chromosome arms in a somatic cell of a particular species.
gamete A germ cell having a haploid chromosomal complement. Gametes from parents of opposite sexes fuse to form zygotes.
gametic selection The forces acting to cause differential reproductive success of one allele over another in a heterozygote.
gametophyte The haploid stage of a plant life cycle that produces gametes (by mitosis). It alternates with a diploid, sporophyte generation.
G-bands Eukaryotic chromosomal bands produced by treatment with Giemsa stain.
gene Inherited determinant of the phenotype. *See* cistron; locus.
gene amplification A process or processes by which the cell increases the number of repeats of a particular gene within the genome.
gene cloning Production of large numbers of a piece of DNA after that piece of DNA is inserted into a vector and taken up by a cell. Cloning occurs as the vector replicates.
gene conversion In Ascomycete fungi, a 2:2 ratio of alleles is expected after meiosis, yet a 3:1 ratio is sometimes observed. The gene conversion mechanism is explained by repair of heteroduplex DNA produced by recombination.
gene family A group of genes that has arisen by duplication of an ancestral gene. The genes in the family may or may not have diverged.
gene flow The movement of genes from one population to another by interbreeding between individuals in the two populations.
gene pool All of the alleles available among the reproductive members of a population from which gametes can be drawn.
generalized transduction Form of transduction in which any region of the host genome can be transduced. *See* specialized transduction.
general transcription factors Eukaryotic proteins that form part of the RNA polymerase holoenzymes.
genetic code The linear sequences of nucleotides that specify the amino acids during the process of translation at the ribosome.
genetic engineering Popular term for recombinant DNA technology. *See* recombinant DNA technology.
genetic load The relative decrease in the mean fitness of a population due to the presence of genotypes that have less than the highest fitness.
genetic polymorphism The occurrence together in the same population of more than one allele at the same locus, with the least frequent allele occurring more frequently than can be accounted for by mutation.
genic balance theory Bridges's theory that the sex of a fruit fly is determined by the relative number of X chromosomes and autosomal sets.
genome The entire genetic complement of a prokaryote or virus or the haploid genetic complement of a eukaryote.
genomic equivalence The concept that differentiated cells in a eukaryotic organism have identical genetic contents.
genomic library A set of cloned fragments making up the entire genome of an organism or species.
genomics The study of the mapping and sequencing of genomes. Bioinformatics is the science of mining the data from these DNA sequences obtained from sequencing.
genophore The chromosome (genetic material) of prokaryotes and viruses.
genotype The genes that an organism possesses.
Giemsa stain A complex of stains specific for the phosphate groups of DNA.
Goldstein-Hogness box *See* TATA box.
green fluorescent protein A reporter system that uses the gene from a jellyfish that specifies a protein that fluoresces green when ultraviolet light is shined on it, indicating the success of a transfection experiment.
group I introns Self-splicing introns that require a guanine-containing nucleotide for splicing; the intron is released in a linear form.
group II introns Self-splicing introns that do not require an external nucleotide for splicing; the intron is released in a lariat form.
group selection Selection for traits that would be beneficial to a population at the expense of the individual possessing the trait.
G-tetraplex A structure of four guanines that can base pair to form a planar structure that may be involved in novel structures at the end of eukaryotic chromosomes.
guanine *See* purines.
guide RNA (gRNA) RNA that guides the insertion of uridines (RNA editing) into messenger RNAs in trypanosomes. Found in transcripts from minicircles and maxicircles of DNA in kinetoplasts.
gynandromorphs Mosaic individuals having simultaneous aspects of both the male and the female phenotype.

hammerhead ribozyme A catalytic RNA, shaped like a hammerhead, capable of splitting other RNA molecules with appropriate complementary sequences.
haplodiploidy The sex-determining mechanism found in some insect groups among which males are haploid and females are diploid.
haploid The state of having one copy of each chromosome per nucleus or cell.
HAT medium A selection medium for hybrid cell lines; contains hypoxanthine, aminopterin, and thymidine. $HPRT^+$ TK^+ cell lines can survive in this medium.
heat shock proteins Proteins that appear in a cell after the cell has been subjected to elevated temperatures.
helicase A protein that unwinds DNA, usually at replicating Y-junctions.
helix-turn-helix motif Configuration found in DNA-binding proteins consisting of a recognition helix and a stabilizing helix. separated by a short turn.
hemizygous The condition of loci present in only one copy in a diploid organism, such as loci on the X chromosome of the heterogametic sex of a diploid species.
heritability A measure of the degree to which the variance in the distribution of a phenotype is due to genetic causes. In the broad sense, it is measured by the total genetic variance divided by the total phenotypic variance. In the narrow sense, it is measured by the genetic variance due to additive genes divided by the total phenotypic variance.
hermaphrodite An individual with both male and female genitalia.
heterochromatin Chromatin that remains tightly coiled (and darkly staining) throughout the cell cycle.
heteroduplex DNA *See* hybrid DNA.
heterogametic The sex with heteromorphic sex chromosomes; during meiosis, it produces different kinds of gametes in accordance with these sex chromosomes.
heterogeneous nuclear mRNA (hnRNA) The original RNA transcripts found in eukaryotic nuclei before posttranscriptional modifications.
heterokaryon A cell that contains two or more nuclei from different origins.
heteromorphic chromosome pair Members of a homologous pair of chromosomes that are not morphologically identical (e.g., the sex chromosomes).
heteroplasmy The existence within an organism of genetic heterogeneity within the populations of mitochondria or chloroplasts.
heterothallic A botanical term used for organisms in which the two sexes reside in different individuals.
heterotrophs Organisms that require an organic form of carbon as a carbon source.
heterozygote A diploid or polyploid with different alleles at a particular locus.
heterozygote advantage A selection model in which heterozygotes have the highest fitness.
heterozygous DNA *See* hybrid DNA.
Hfr High frequency of recombination. A strain of bacteria that has incorporated an F factor into its chromosome and can then transfer the chromosome during conjugation.
histone acetyl transferases (HATs) Proteins that remodel chromatin by acetylating histones.
histones Arginine- and lysine-rich basic proteins making up a substantial portion of eukaryotic nucleoprotein.
hnRNA *See* heterogeneous nuclear mRNA.
Hogness box *See* TATA box.
holandric trait Trait controlled by a locus found only on the Y chromosome. Involves father-to-son transmission.
Holliday junction A junction point between two cross-linked DNA double helices. It is an intermediate stage in DNA recombination.
holoenzyme The complete enzyme, including all subunits. Often used in reference to RNA and DNA polymerases.
homeo box A consensus sequence of about 180 base pairs discovered in homeotic genes in *Drosophila*. Also found in other developmentally important genes from yeast to human beings.
homeo domain The sixty amino acid polypeptide translated from the homeo box.
homeotic gene Gene that controls the developmental fate of a cell type; mutations of the homeotic gene cause one cell type to follow the developmental pathway of another cell type.
homogametic The sex with homomorphic sex chromosomes; it produces only one kind of gamete in regard to the sex chromosomes.
homologous chromosomes Members of a pair of essentially identical chromosomes that synapse during meiosis.
homologous recombination Breakage and reunion between homologous lengths of DNA mediated by RecA and RecBCD.
homomorphic chromosome pairs Morphologically identical members of a homologous pair of chromosomes.
homoplasmy The existence within an organism of only one type of plastid; usually referring to the genetic identity of mitochondria or chloroplasts.
homothallic A botanical term used for groups whose individuals are not of different sexes.
homozygote A diploid or a polyploid with identical alleles at a locus.
humoral immunity Immunity due to antibodies in the serum and lymph.
H-Y antigen The histocompatibility Y-antigen, a protein found on the cell surfaces of male mammals.
hybrid Offspring of unlike parents.
hybrid DNA DNA whose two strands have different origins.
hybridoma A cell resulting from the fusion of a spleen cell and a multiple myeloma cell. These cells can be maintained indefinitely in cell culture, in which they produce monoclonal antibodies.
hybrid plasmid A plasmid that contains an inserted piece of foreign DNA.
hybrid vector *See* hybrid vehicle.
hybrid vehicle A plasmid or phage containing an inserted piece of foreign DNA.
hybrid zone Geographical region in which previously isolated populations that have evolved differences come into contact and form hybrids.
hyperplasia Excessive cell growth that does not involve pathological changes to the cells.
hypervariable loci Loci with many alleles; especially those whose variation is due to variable numbers of tandem repeats.
hypostatic gene A gene whose expression is masked by an epistatic gene.
identity by descent The state of two alleles when they are identical copies of the same ancestral allele (autozygous).
idiogram A photograph or diagram of the chromosomes of a cell arranged in an orderly fashion. *See* karyotype.
idiotypic variation Variation in the variable parts of immunoglobulin genes.
idling reaction The production of guanosine tetraphosphate (3′-ppGpp-5′) by the stringent factor when a ribosome encounters an uncharged transfer RNA in the A site.
immunity The ability of an organism to resist infection.
immunoglobulins (Igs) Specific proteins produced by derivatives of B lymphocytes that protect an organism from antigens.
imprinting *See* molecular imprinting.
imprinting center (IC) A region responsible for the control of imprinting. The imprinting mark is almost certainly DNA methylation, which is able to turn off gene transcription.
inbreeding The mating of genetically related individuals.
inbreeding coefficient, F The probability of autozygosity.
inbreeding depression A depression of vigor or yield due to inbreeding.
incestuous A mating between blood relatives who are more closely related than the law of the land allows.
inclusive fitness The expansion of the concept of the fitness of a genotype to include benefits accrued to relatives of an individual since relatives share parts of their genomes. An apparently altruistic act toward a relative may thus enhance the fitness of the individual performing the act.

incomplete dominance The situation in which both alleles of the heterozygote influence the phenotype. The phenotype is usually intermediate between the two homozygous forms.

independent assortment, rule of Mendel's second rule, describing the independent segregation of alleles of different loci.

inducible system A system (a coordinated group of enzymes involved in a catabolic pathway) is inducible if the metabolite it works upon causes transcription of the genes controlling these enzymes. These systems are primarily prokaryotic operons.

induction In regard to temperate phages, the process of causing a prophage to become virulent.

informatics *See* bioinformatics.

initiation codon The messenger RNA sequence AUG, which specifies methionine, the first amino acid used in the translation process. (Occasionally, GUG is recognized as an initiation codon.)

initiation complex The complex formed for initiation of translation. It consists of the 30S ribosomal subunit, messenger RNA, N-formyl-methionine transfer RNA, and three initiation factors.

initiation factors (IFI, IF2, IF3) Proteins (prokaryotic with eukaryotic analogues) required for the proper initiation of translation.

initiator element (inr) A CT-rich area found in RNA polymerase II promoters without TATA boxes.

initiator proteins Proteins that recognize the origin of replication on a replicon and take part in primosome construction.

insertion mutagenesis Change in gene action due to an insertion event that either changes a gene directly or disrupts control mechanisms.

insertion sequences (IS) Small, simple transposons. *See* transposable genetic element.

inside marker The middle locus of three linked loci.

intercalary heterochromatin Heterochromatin, other than centromeric heterochromatin, dispersed throughout eukaryotic chromosomes.

intergenic suppression A mutation at a second locus that apparently restores the wild-type phenotype to a mutant at a first locus.

interkinesis The abbreviated interphase that occurs between meiosis I and II. No DNA replication occurs here.

internal ribosome entry site Sequence in eukaryotic messenger RNAs that allows ribosomes to initiate translation at a point other than the 5′ cap.

interphase The metabolically active, nondividing stage of the cell cycle.

interpolar microtubules Microtubules extending from one pole of the spindle and overlapping spindle fibers from the other pole, but not in contact with kinetochores.

interrupted mating A mapping technique in which bacterial conjugation is disrupted after specified time intervals.

intersex An organism with external sexual characteristics of both sexes.

intervening sequences (introns) DNA sequences within a gene that are transcribed but removed prior to translation.

intra-allelic complementation The restoration of activity in an enzyme made of subunits in a heterozygote of two mutants that, when homozygous, are not active. Caused by the interaction of the subunits in the protein.

intragenic suppression A second change within a mutant gene that results in an apparent restoration of the original phenotype.

intron *See* intervening sequences.

inversion The replacement of a section of a chromosome in the reverse orientation.

inverted repeat sequence A nucleotide sequence read in opposite orientations on the same double helix.

in vitro Biological or chemical work done in the test tube (literally, "in glass") rather than in living systems.

IS elements *See* insertion sequences.

isochromosome A chromosome with two genetically and morphologically identical arms.

isozymes Different electrophoretic forms of the same enzyme. Unlike allozymes, isozymes are due to differing subunit configurations rather than allelic differences.

junctional diversity Variability in immunoglobulins caused by variation in the exact crossover point during V-J, V-D, and D-J joining.

kappa particles The bacterialike particles that give a *Paramecium* the killer phenotype.

karyokinesis The process of nuclear division. *See* cytokinesis.

karyotype The chromosome complement of a cell. *See* idiogram.

kinesin Microtubule motor protein.

kinetochore The chromosomal attachment point for the spindle fibers, located on the centromere.

kinetochore microtubules Microtubules radiating from the centrosome and attached to kinetochores of chromosomes during mitosis and meiosis.

kin selection The mode of natural selection that acts on an individual's inclusive fitness.

Klenow fragment Proteolytic fragment—obtained by treatment with a protease, or protein-cleaving enzyme—of *E. coli* DNA polymerase I with both $5'\rightarrow3'$ polymerase activity and $3'\rightarrow5'$ exonuclease activity. (It has been studied extensively because it has been easier for X-ray crystallographers and biochemists to work with this fragment than with the whole enzyme.)

knockout mice Transgenic mice that have been made homozygous for a nonfunctioning allele at a particular locus.

***lac* operon** The inducible operon, including three loci involved in the uptake and breakdown of lactose.

ladder gel *See* stepladder gel.

lagging strand Strand of DNA being replicated discontinuously.

lampbrush chromosomes Chromosomes of amphibian oocytes having loops that are suggestive of a lampbrush.

leader The length of messenger RNA from the 5′ end to the initiation codon, AUG.

leader peptide gene A small gene within the attenuator control region of a repressible amino acid operon. Translation of the gene tests the concentration of amino acids in the cell.

leader transcript The messenger RNA transcribed by the attenuator region of a repressible amino acid operon. The transcript is capable of several alternative stem-loop structures, depending on the translation of a short leader peptide gene.

leading strand Strand of DNA being replicated continuously.

leptonema (leptotene stage) The first stage of prophase I of meiosis, in which chromosomes become distinct.

lethal-equivalent alleles Alleles whose summed effect is that of lethality—for example, four alleles, each of which would be lethal 25% of the time (or to 25% of their bearers), are equivalent to one lethal allele.

leucine zipper Configuration of a DNA-binding protein in which leucine residues on two helices interdigitate, in zipper fashion, to stabilize the protein.

leukemia Cancer of the bone marrow resulting in excess production of leukocytes.

level of significance The probability value in statistics used to reject the null hypothesis.

linkage The association of loci on the same chromosome.

linkage disequilibrium The condition among alleles at different loci such that allelic combinations in a gamete do not follow the product rule of probability.

linkage equilibrium The condition among alleles at different loci such that any allelic combination in a gamete occurs as the product of the frequencies of each allele at its own locus.

linkage groups Associations of loci on the same chromosome. In a species, there are as many linkage groups as there are homologous pairs of chromosomes.
linkage number The number of times one strand of a helix coils about the other.
linker A small segment of DNA that contains a restriction site. It can be added to blunt-ended DNA to give that DNA a particular restriction site for cloning.
liposomes Transfecting DNA is delivered to target cells by way of these membrane-bound vesicles.
locus The position of a gene on a chromosome (plural: *loci*).
***lod* score method** A technique (*l*ogarithmic *od*ds) for determining the most likely recombination frequency between two loci from pedigree data.
long interspersed elements (LINEs) Sequences of DNA, up to seven thousand base pairs in length, interspersed in eukaryotic chromosomes in many copies.
lymphoma Cancer of the lymph nodes and spleen that causes excessive production of lymphocytes.
Lyon hypothesis The hypothesis that suggests that the Barr body is an inactivated X chromosome.
lysate The contents released from a lysed cell.
lysis The breaking open of a cell by the destruction of its wall or membrane.
lysogenic The state of a bacterial cell that has an integrated phage (prophage) in its chromosome.
major histocompatibility complex A group of highly polymorphic genes whose products appear on the surfaces of cells, imparting to them the property of "self" (belonging to that organism). Some other functions are also involved.
mapping The process of locating the positions of genes on chromosomes.
mapping function The mathematical relationship between measured map distance in a given experiment and the actual recombination frequency.
map unit The distance equal to 1% recombination between two loci.
mate-killer A *Paramecium* phenotype induced by intracellular bacterialike mu particles.
maternal effect The effect of the maternal parent's genotype on the phenotype of her offspring.
maternal-effect gene A gene expressed in maternal tissue that influences a developing embryo.
mating type In many species of microorganisms, individuals can be divided into two mating types. Mating can take place only between individuals of opposite mating types due to the interaction of cell surface components.
maturation-promoting factor (MPF) A protein complex of cyclin B and p34*cdc2* that initiates mitosis during the cell cycle. Also called the mitosis-promoting factor.
mean The arithmetic average; the sum of the data divided by the sample size.
mean fitness of the population, $\overline{W}$ The sum of the fitnesses of the genotypes of a population weighted by their proportions; hence, a weighted mean fitness.
meiosis The nuclear process in diploid eukaryotes that results in gametes or spores with only one member of each original homologous pair of chromosomes per nucleus.
meiotic drive *See* gametic selection.
merozygote A bacterial cell having a second copy of a particular chromosomal region in the form of an exogenote.
messenger RNA (mRNA) A complementary copy of a gene that is translated into a polypeptide at the ribosome.
metacentric chromosome A chromosome whose centromere is located in the middle.
metafemale A fruit fly with an X/A ratio greater than unity.
metagon An RNA necessary for the maintenance of mu particles in *Paramecium*.
metamale A fruit fly with X/A ratio below 0.5.
metaphase The stage of mitosis or meiosis in which spindle fibers are attached to kinetochores, and the chromosomes are positioned in the equatorial plane of the cell.
metaphase plate The plane of the equator of the spindle into which chromosomes are positioned during metaphase.
metastasis The migration of cancerous cells to other parts of the body.
metrical variation *See* continuous variation.
microsatellite DNA Repeats of very short sequences of DNA, such as CACACACA, dispersed throughout the eukaryotic genome. The loci can be studied by polymerase chain reaction amplification.
microtubule organizing center Active center from which microtubules are organized. The spindle is organized by the centrosome, which may or may not contain a centriole.
microtubules Hollow cylinders made of the protein tubulin (α and β subunits) that make up, among other things, the spindle fibers.
mimicry A phenomenon in which an individual gains an advantage by looking like the individuals of a different species.
minimal medium A culture medium for microorganisms that contains the minimal necessities for growth of the wild-type.
mismatch repair A form of excision repair initiated at the sites of mismatched bases in DNA.
missense mutation Mutations that change a codon for one amino acid into a codon for a different amino acid.
mitochondrion The eukaryotic cellular organelle in which the Krebs cycle and electron transport reactions take place.
mitosis The nuclear division producing two daughter nuclei identical to the original nucleus.
mitosis-promoting factor *See* maturation-promoting factor (MPF).
mitotic apparatus *See* spindle.
mixed families Groups of four codons sharing their first two bases and coding for more than one amino acid.
modern linkage map A chromosomal map based on the positions of RFLP markers along its length.
molecular chaperone A protein that aids in the folding of a second protein. The chaperone prevents proteins from forming structures that would be inactive.
molecular evolutionary clock A measurement of evolutionary time in nucleotide substitutions per year.
molecular imprinting The phenomenon in which there is differential expression of a gene depending on whether it was maternally or paternally inherited.
molecular mimicry The situation in which one type of molecule resembles another type in order to function. For example, the prokaryotic ribosomal release factors, RF1 and RF2, mimic the structure of a transfer RNA.
monocistronic Usually referring to a messenger RNA that carries the information for only one gene (cistron).
monoclonal antibody The antibody from a clone of cells producing the same antibody. An individual with multiple myeloma usually produces monoclonal antibodies.
monohybrids Offspring of parents that differ in only one genetic characteristic. Usually implies heterozygosity at a single locus under study.
monosomic A diploid cell missing a single chromosome.
monovalent A single chromosome composed of two sister chromatids. Equivalent to a dyad.
morphogen A substance transported into or produced in a developing embryo that diffuses to form a gradient that helps determine cell differentiation.
morphological species concept The idea that organisms are classified in the same species if they appear similar.
mosaicism The condition of being a mosaic. *See* mosaics.
mosaics Individuals made up of two or more cell lines in which the cells originated in the same zygote.
mRNA *See* messenger RNA (mRNA).
Müllerian mimicry A form of mimicry in which noxious species evolve to resemble each other.
multihybrid An organism heterozygous at numerous loci.

multinomial expansion The terms generated when a multinomial is raised to a power.
mu particles Bacterialike particles found in the cytoplasm of *Paramecium* that impart the mate-killer phenotype.
mutability The ability to change.
mutants Alternative phenotypes to the wild-type; the phenotypes produced by alternative alleles.
mutation The process by which a gene or chromosome changes structurally; the end result of this process.
mutation rate The proportion of mutations per cell division in bacteria or single-celled organisms or the proportion of mutations per gamete in higher organisms.
mutator mutations Mutations of DNA polymerase that increase the overall mutation rate of a cell or of an organism.
muton A term Benzer coined for the smallest mutable site within a cistron.
natural selection The process in nature whereby one genotype leaves more offspring than another genotype because of superior life history attributes such as survival or fecundity.
negative interference The phenomenon whereby a crossover in a particular region facilitates the occurrence of other apparent crossovers in the same region of the chromosome.
N-end rule The life span of a protein is determined by its amino-terminal (N-terminal) amino acid.
neo-Darwinism The merger of classical Darwinian evolution with population genetics.
neoplasm New growth of abnormal tissue.
neutral gene hypothesis The hypothesis that most genetic variation in natural populations is not maintained by natural selection.
NF *See* fundamental number.
nickase *See* DNA gyrase.
noncoding strand *See* anticoding strand.
nondisjunction The failure of a pair of homologous chromosomes to separate properly during meiosis.
nonhistone proteins The proteins remaining in chromatin after the histones are removed.
nonparental ditype (NPD) A spore arrangement in Ascomycete fungi that contains only the two recombinant-type ascospores (assuming two segregating loci).
nonparentals *See* recombinants.
nonrecombinants In mapping studies, offspring that have alleles arranged as in the original parents.
nonsense codon One of the messenger RNA sequences (UAA, UAG, UGA) that signals the termination of translation.
nonsense mutations Mutations that change a codon for an amino acid into a nonsense codon.
normal distribution Any of a family of bell-shaped frequency curves whose relative positions and shapes are defined on the basis of the mean and standard deviation.
northern blotting A gel transfer technique used for RNA. *See* Southern blotting.
N segments Sequences of nucleotides added in a template-free fashion at the joining junctions of heavy-chain antibody genes.
nuclease-hypersensitive site A region of a eukaryotic chromosome that is specifically vulnerable to nuclease attack because it is not wrapped as nucleosomes.
nucleolar organizer The chromosomal region around which the nucleolus forms; site of tandem repeats of the major ribosomal RNA gene.
nucleolus The globular, nuclear organelle formed at the nucleolar organizer. Site of ribosome construction.
nucleoprotein The substance of eukaryotic chromosomes consisting of proteins and nucleic acids.
nucleoside A sugar-base compound that is a nucleotide precursor. Nucleotides are nucleoside phosphates.
nucleosomes Arrangements of DNA and histones forming regular spherical structures in eukaryotic chromatin.
nucleotide Subunits that polymerize into nucleic acids (DNA or RNA). Each nucleotide consists of a nitrogenous base, a sugar, and one or more phosphate groups.
nucleotide excision repair The DNA excision repair mechanism responsible for repairing thymine dimers and other lesions. Enzymes excise a short segment of one of the DNA strands and then repair and ligate the DNA.
null hypothesis The statistical hypothesis that there are no differences between observed data and those data expected based on the assumption of no experimental effect.
nullisomic A diploid cell missing both copies of the same chromosome.
numerator elements Genes on the X chromosome in *Drosophila* that regulate the sex switch gene (*sxl*) to the on condition (femaleness). Refers to the numerator of the X/A genic balance equation.
nutritional-requirement mutants *See* auxotrophs.
Okazaki fragments Segments of newly replicated DNA produced during discontinuous DNA replication.
oncogene Genes capable of transforming a cell. They are found in the active state in retroviruses and transformed cells and in the inactive state in nontransformed cells, in which they are called proto-oncogenes.
one-gene-one-enzyme hypothesis Hypothesis of Beadle and Tatum that states that one gene controls the production of one enzyme. Later modified to the concept that one cistron controls the production of one polypeptide.
oogenesis The process of ovum formation in female animals.
oogonia Cells in females that produce primary oocytes by mitosis.
open reading frames (ORFs) Sequence of codons between the initiation and termination codons in a gene.
operator A DNA sequence recognized by a repressor protein or repressor-corepressor complex. When the operator is complexed with the repressor, transcription is prevented.
operon A sequence of adjacent genes all under the transcriptional control of the same operator.
origin recognition complex (ORC) A complex of six proteins that bind to the eukaryotic autonomously replicating sequences (ARS). Needed for the initiation of DNA replication in concert with other proteins.
outbreeding The mating of genetically unrelated individuals.
ovum Egg. The one functional product of each meiosis in female animals.
pachynema (pachytene stage) The stage of prophase I of meiosis in which chromatids are first distinctly visible.
palindrome A sequence of words, phrases, or nucleotides that reads the same regardless of the direction from which one starts; the sites of recognition of type II restriction endonucleases.
panmictic Referring to unstructured (random mating) populations.
paracentric inversion A chromosomal inversion that does not include the centromere.
paramecin A toxin liberated by a "killer" *Paramecium*.
parameters Measurements of attributes of a population; denoted by Greek letters.
parapatric speciation Speciation in which reproductive isolating mechanisms evolve when a population enters a new niche or habitat within the range of the parent species.
parasegment The first series of segments that form in a developing insect embryo; they form after about 5.5 hours in the developing *Drosophila* embryo.
parental ditype (PD) A spore arrangement in Ascomycete fungi that contains only the two nonrecombinant-type ascospores.
parental imprinting *See* molecular imprinting.
parentals *See* nonrecombinants.
parthenogenesis The development of an individual from an unfertilized egg that did not arise by meiotic chromosomal reduction.
partial digest A restriction digest that has not been allowed to go to completion, and thus contains pieces of DNA that have restriction endonuclease sites that have not been cleaved.

partial dominance *See* incomplete dominance.
Pascal's triangle A triangular array of numbers made up of the coefficients of the binomial expansion.
path diagram A modified pedigree showing only the direct line of descent from common ancestors.
pedigree A representation of the ancestry of an individual or family; a family tree.
penetrance The normal appearance of genetically controlled traits in the phenotype.
peptidyl transferase The enzymatic center responsible for peptide bond formation during translation at the ribosome.
pericentric inversion A chromosomal inversion that includes the centromere.
permissive temperature A temperature at which temperature-sensitive mutants are normal.
PEST hypothesis Degradation of a protein in less than two hours is signaled by a region within the protein rich in proline (P), glutamic acid (E), serine (S), or threonine (T).
petite mutations Mutations of yeast that produce small colonies, like those grown under anaerobic conditions.
phages *See* bacteriophages.
phenocopy A phenotype that is not genetically controlled but looks like a genetically controlled phenotype.
phenotype The observable attributes of an organism.
pheromone A chemical signal, analogous to a hormone, that passes information between individuals.
phosphodiester bond A diester bond linking two nucleotides together (between phosphoric acid and sugars) to form the nucleotide polymers DNA and RNA.
photocrosslinking A technique used to determine which moieties (proteins, DNA) are in close proximity during a particular process.
photoreactivation The process whereby dimerized pyrimidines (usually thymines) in DNA are restored by an enzyme (deoxyribodipyrimidine photolyase) that requires light energy.
phyletic evolution *See* anagenesis.
phyletic gradualism The process of gradual evolutionary change over time.
phylogenetic tree A diagram showing evolutionary lineages of organisms.
physical map Chromosomal map in which distances are in physical units of base pairs. These maps can be of microsatellite markers or of sequence-tagged sites.
pili (fimbriae) Hairlike projections on the surface of bacteria; Latin for "hair."
plaques Clear areas on a bacterial lawn caused by cell lysis due to viral attack.
plasmid An autonomous, self-replicating genetic particle, usually of double-stranded DNA.
plastid A chloroplast prior to the development of chlorophyll.
pleiotropy The phenomenon whereby a single mutant affects several apparently unrelated aspects of the phenotype.
point centromere The type of centromere, such as that found in *Saccharomyces cerevisiae,* that has defined sequences large enough to accommodate one spindle microtubule.
point mutations Small mutations that consist of a replacement, addition, or deletion of one or a few bases.
polar bodies The small cells that are the by-products of meiosis in female animals. One functional ovum and as many as three polar bodies result from meiosis of each primary oocyte.
polarity Meaning "directionality" and referring either to an effect seen in only one direction from a point of origin or to the fact that linear entities (such as a single strand of DNA) have ends that differ from each other.
polar mutant An organism with a mutation, usually within an operon, that prevents the expression of genes distal to itself.
pollen grain The male gametophyte in higher plants.
poly-A tail A sequence of adenosine nucleotides added to the 3′ end of eukaryotic messenger RNAs.
polycistronic Referring to prokaryotic messenger RNAs that contain several genes within the same messenger RNA transcript.
polygenic inheritance *See* quantitative inheritance.
polymerase chain reaction (PCR) A method to amplify DNA segments rapidly in temperature-controlled cycles of denaturation, primer binding, and replication.
polymerase cycling The process by which a DNA polymerase III enzyme completes an Okazaki fragment, releases it, and begins synthesis of the next Okazaki fragment.
polymerized Formed into a complex compound by linking together smaller elements.
polynucleotide phosphorylase An enzyme that can polymerize diphosphate nucleotides without the need for a primer. The function of this enzyme *in vivo* is probably in its reverse role as an RNA exonuclease.
polyploids Organisms with greater than two chromosome sets.
polyribosome *See* polysome.
polysome The configuration of several ribosomes simultaneously translating the same messenger RNA. Shortened form of the term *polyribosome*.
polytene chromosome Large chromosome, seen, for example, in *Drosophila* salivary glands, consisting of many chromatids formed by rounds of endomitosis. Synapsis of homologous chromosomes occurs during the process.
population A group of organisms of the same species relatively isolated from other groups of the same species. *See* deme.
position effect An alteration of phenotype caused by a change in the relative arrangement of the genetic material.
positive interference When the occurrence of one crossover reduces the probability that a second will occur in the same region.
postreplicative repair A DNA repair process initiated when DNA polymerase bypasses a damaged area.
posttranscriptional modifications Changes in eukaryotic messenger RNA made after transcription has been completed. These changes include additions of caps and tails and removal of introns.
preemptor stem A configuration of leader transcript messenger RNA that does not terminate transcription in the attenuator-controlled amino acid operons.
pre-initiation complex (PIC) The form of the RNA polymerase II enzyme with general transcription factors bound equivalent to the *E. coli* holoenzyme. Phosphorylation of the enzyme then allows transcription to begin.
Pribnow box Consensus sequence of TATAAT in prokaryotic promoters centered at the position −10.
primary oocytes The cells that undergo meiosis in female animals.
primary spermatocytes The cells that undergo meiosis in male animals.
primary structure The sequence of polymerized amino acids in a protein.
primary transcript The product of eukaryotic transcription before posttranscriptional modification takes place.
primase An enzyme that creates a messenger RNA primer for Okazaki fragment initiation.
primer In DNA replication, a length of double-stranded DNA that continues as a single-stranded template in the 3′ to 5′ direction.
primosome A complex of two proteins, a primase and helicase, that initiates RNA primers on the lagging DNA strand during DNA replication.
prion Infectious agent responsible for several neurological diseases (scrapie, kuru, Creutzfeld-Jakob syndrome, mad-cow disease). It is a protein that lacks DNA or RNA.
probability The expectation of the occurrence of a particular event.
probability theory The conceptual framework concerned with quantification of probabilities. *See* probability.
proband *See* propositus.
probe In recombinant DNA work, a radioactive nucleic acid complementary to a region being searched for in a restriction digest or genomic library.
processivity The ability of an enzyme to repetitively continue its catalytic function without dissociating from its substrate.

product rule The rule that states that the probability that two independent events will both occur is the product of their separate probabilities.
progeny testing Breeding of offspring to determine their genotypes and those of their parents.
prokaryotes Organisms that lack true nuclei.
promoter A DNA region that RNA polymerase binds to in order to initiate transcription.
proofread Technically, to read for the purpose of detecting errors for later correction. DNA polymerase has 3′ to 5′ exonuclease activity, which it uses during polymerization to remove incorrect nucleotides it has recently added.
prophage A temperate phage replicating with the host that can later initiate the lytic cycle.
prophase The initial stage of mitosis or meiosis in which chromosomes become visible and the spindle apparatus forms.
proplastid Mutant plastids that do not grow and develop into chloroplasts.
propositus (proposita) The person through whom a pedigree was discovered.
proteasome A barrel-shaped cellular organelle for protein breakdown involving the ubuiquitin pathway.
proteome From *prote*ins of the gen*ome;* the complete set of proteins from a particular genome. It is the protein analogue to "genome."
proteomics The study of the complete set of proteins from a particular genome. It is the protein analogue to "genomics."
proto-oncogene A cellular oncogene in an untransformed cell.
prototrophs Strains of organisms that can survive on minimal medium.
pseudoalleles Genes that are functionally but not structurally allelic. Within gene families, pseudoalleles are alleles that are not expressed.
pseudoautosomal gene A gene that occurs on both sex-determining heteromorphic chromosomes.
pseudodominance The phenomenon in which a recessive allele shows itself in the phenotype when only one copy of the allele is present, as in hemizygous alleles or in deletion heterozygotes.
P (peptidyl) site The site on the ribosome occupied by the peptidyl-tRNA just before peptide bond formation.
punctuated equilibrium The evolutionary process involving long periods without change (stasis) punctuated by short periods of rapid speciation.
Punnett square A diagrammatic representation of a particular cross used to predict the progeny of the cross.
purines Nitrogenous bases of which guanine and adenine are found in DNA and RNA.
pyrimidines Nitrogenous bases of which thymine is found in DNA, uracil in RNA, and cytosine in both.
quantitative inheritance The mechanism of genetic control of traits showing continuous variation.
quantitative trait loci Chromosomal regions contributing to the inheritance of a quantitative trait. These regions may contain one or more polygenes that contribute to the phenotype.
quantitative variation *See* continuous variation.
quaternary structure The association of polypeptide subunits to form the final structure of a protein.
random genetic drift Changes in allelic frequency due to sampling error.
random mating The mating of individuals in a population such that the union of individuals with the trait under study occurs according to the product rule of probability.
random strand analysis Mapping studies in organisms that do not keep all the products of meiosis together.
read-through Transcription or translation beyond the normal termination signals in DNA or RNA, respectively.
realized heritability Heritability measured by a response to selection.
recessive An allele (or phenotype) that does not express itself in the heterozygous condition.
reciprocal cross A cross with the phenotype of each sex reversed as compared with the original cross. Made to test the role of parental sex on inheritance pattern.
reciprocal translocation A chromosomal configuration in which the ends of two nonhomologous chromosomes are broken off and become attached to the nonhomologues.
recombinant DNA technology Techniques of gene cloning. Recombinant DNA refers to the hybrid of foreign and vector DNA. *See* gene cloning.
recombinant plasmid A plasmid that contains an inserted piece of foreign DNA.
recombinants In mapping studies, offspring with allelic arrangements made up of a combination of the original parental alleles.
recombination The nonparental arrangement of alleles in progeny that can result from either independent assortment or crossing over.
recombination nodule Proteinaceous nodules found on bivalents during zygonema and pachynema associated with crossing over.
recon A term Benzer coined for the smallest recombinable unit within a cistron.
reductional division The first meiotic division. It reduces the number of chromosomes and centromeres to half that in the original cell.
regional centromere The type of centromere found in higher eukaryotes that can accommodate several spindle microtubules.
regulator gene A gene primarily involved in control of the production of another gene's product.
reinitiation The initiation of translation by a ribosome that has just completed translation of a region of the messenger RNA upstream of the current point of initiation.
relative Darwinian fitness *See* fitness.
relaxed mutant A mutant that does not exhibit the stringent response under amino acid starvation.
release factors (RFl and RF2) Proteins in prokaryotes responsible for termination of translation and release of the newly synthesized polypeptide when a nonsense codon appears in the A site of the ribosome. Replaced by eRF in eukaryotes.
repetitive DNA DNA made up of copies of the same nucleotide sequence.
replica-plating A technique to rapidly transfer microorganism colonies to numerous petri plates.
replication-coupling assembly factor A protein complex in fruit flies that assembles new nucleosomes.
replicons A replicating genetic unit including a length of DNA and its site for the initiation of replication.
replisome The DNA-replicating structure at the Y-junction, consisting of two DNA polymerase III enzymes and a primosome (primase and DNA helicase).
reporter systems Genetic constructs that allow an investigator to determine that a specific locus is active by measuring the phenotypic expression of an associated locus, such as the luciferase reporter, which glows if watered with luciferin.
repressible system A coordinated group of enzymes involved in a synthetic pathway (anabolic) is repressible if excess quantities of the end product of the pathway lead to the termination of transcription of the genes for the enzymes. These systems are primarily prokaryotic operons.
repressor The protein product of a regulator gene that acts to control transcription of inducible and repressible operons.
reproductive isolating mechanisms Environmental, behavioral, mechanical, and physiological barriers that prevent two individuals of different populations from producing viable progeny.
reproductive success The relative production of offspring by a particular genotype.
repulsion Allelic arrangement in which each homologous chromosome has mutant and wild-type alleles.
resistance transfer factor Infectious transfer part of R plasmids.

restricted transduction *See* specialized transduction.
restriction digest The results of the action of a restriction endonuclease on a DNA sample.
restriction endonucleases Endonucleases that recognize certain DNA sequences, then cleave them. They protect cells from viral infection; they are useful in recombinant DNA work.
restriction fragment length polymorphism (RFLP) Variations (among individuals) in banding patterns of electrophoresed restriction digests.
restriction map A physical map of a piece of DNA showing recognition sites of specific restriction endonucleases separated by lengths marked in numbers of bases.
restriction site The sequence of DNA recognized by a restriction endonuclease.
restrictive temperature A temperature at which temperature-sensitive mutants display the mutant phenotype.
retinoblastoma A childhood cancer of retinoblast cells caused by the inactivation of an anti-oncogene.
retrotransposons Transposable genetic elements found in eukaryotic DNA that move through the reverse transcription of an RNA intermediate.
reverse transcriptase An enzyme that can use RNA as a template to synthesize DNA.
reversion The return of a mutant to the wild-type phenotype by way of a second mutational event.
R factors *See* R plasmids.
rho-dependent terminator A DNA sequence signaling the termination of transcription; termination requires the presence of the rho protein.
rho-independent terminator A DNA sequence signaling the termination of transcription; the rho protein is not required for termination.
rho protein A protein involved in the termination of transcription.
ribosomal RNA (rRNA) RNA components of the subunits of the ribosomes.
ribosome recycling factor (RRF) A protein needed to prepare ribosomal subunits that have just finished translating a messenger RNA for another cycle of translation.
ribosomes Organelles at which translation takes place. They are made up of two subunits consisting of RNA and proteins.
ribozyme Catalytic or autocatalytic RNA.
RNA editing The insertion of uridines into messenger RNAs after transcription is completed; controlled by guide RNA. May also involve insertion of cytidines in some organisms or possible deletions of bases.
RNA phages Phages whose genetic material is RNA. They are the simplest phages known.
RNA polymerase The enzyme that polymerizes RNA by using DNA as a template. (Also known as *transcriptase* or *RNA transcriptase.*)
RNA replicase A polymerase enzyme that catalyzes the self-replication of single-stranded RNA.
Robertsonian fusion Fusion of two acrocentric chromosomes at the centromere.
rolling-circle replication A model of DNA replication that accounts for a circular DNA molecule producing linear daughter double helices.
R plasmids Plasmids that carry genes that control resistance to various drugs.
rRNA *See* ribosomal RNA (rRNA).
rule of independent assortment *See* independent assortment, rule of.
rule of segregation *See* segregation, rule of.
sampling distribution The distribution of frequencies with which various possible events could occur, or a probability distribution defined by a particular mathematical expression.
sarcoma Tumor of tissue of mesodermal origin (e.g., muscle, bone, cartilage).
satellite DNA Highly repetitive eukaryotic DNA primarily located around centromeres. Satellite DNA usually has a different buoyant density than the rest of the cell's DNA.
scaffold The eukaryotic chromosomal structure that remains when DNA and histones have been removed.
scanning hypothesis Proposed mechanism by which the eukaryotic ribosome recognizes the initiation region of a messenger RNA after binding the 5′ capped end of it. The ribosome scans the messenger RNA for the initiation codon.
scientific method A procedure scientists use to test hypotheses, making predictions about the outcome of an observation or experiment before the experiment is performed. The results provide support or refutation of the hypothesis.
screening technique A technique to isolate a specific genotype or phenotype of an organism.
secondary oocytes The cells formed by meiosis I in female animals.
secondary spermatoctyes The products of the first meiotic division in male animals.
secondary structure The flat or helical configuration of the polypeptide backbone of a protein.
second-division segregation (SDS) The allelic arrangement in the spores of Ascomycete fungi with ordered spores that indicates a crossover between a locus and its centromere
securin An inhibitory protein that prevents separin from acting on cohesin to separate sister chromatids.
segmentation genes Genes of developing embryos that determine the number and fate of segments.
segregation, rule of Mendel's first principle, which describes how genes pass from one generation to the next.
segregational load Genetic load caused when a population is segregating less fit homozygotes because of heterozygote advantage.
segregation distortion *See* gametic selection.
selection *See* natural selection.
selection coefficients, *s, t* The sum of forces acting to lower the relative reproductive success of a genotype.
selection-mutation equilibrium An equilibrium allelic frequency resulting from the balance between selection against an allele and mutation re-creating this allele.
selective medium A culture medium enriched with a particular substance to allow the growth of particular strains of organisms.
selfed *See* self-fertilization.
self-fertilization Fertilization in which the two gametes come from the same individual.
selfish DNA A segment of the genome with no apparent function, although it can control its own copy number.
semiconservative replication The mode by which DNA replicates. Each strand acts as a template for a new double helix. *See* template.
semisterility Nonviability of a proportion of gametes or zygotes.
sense strand *See* coding strand.
separin An enzyme that breaks down cohesin and allows sister chromatids to separate at the start of anaphase of mitosis.
sequence-tagged sites (STSs) DNA lengths of 100–500 base pairs that are unique in the genome. They are created by polymerase chain reaction amplification of primers that are then tested to be sure the sequence is unique.
sex chromosomes Heteromorphic chromosomes whose distribution in a zygote determines the sex of the organism.
sex-conditioned traits Traits that appear more often in one sex than in another.
sex-determining region Y (*SRY*) The sex switch, or testis-determining factor, in human beings, located on the Y chromosome (*Sry* in mice).
sexduction A process whereby a bacterium gains access to and incorporates foreign DNA brought in by a modified F factor during conjugation.
sex-influenced traits *See* sex-conditioned traits.
Sex-lethal A gene in *Drosophila*, located on the X chromosome, that is a sex switch, directing development toward femaleness when in the "on" state. It is regulated by numerator and denominator elements that act to influence the genic balance ratio (X/A).
sex-limited traits Traits expressed in only one sex. They may be controlled by sex-linked or autosomal loci.

sex-linked The inheritance pattern of loci located on the sex chromosomes (usually the X chromosome in XY species); also refers to the loci themselves.

sex-ratio phenotype A trait in *Drosophila* whereby females produce mostly, if not only, daughters.

sex switch A gene in mammals, normally found on the Y chromosome, that directs the indeterminate gonads towards development as testes.

sexual selection The forces, determined by mate choice, that act to cause one genotype to mate more frequently than another genotype.

Shine-Dalgarno hypothesis A proposal that prokaryotic messenger RNA is aligned at the ribosome by complementarity between the messenger RNA upstream from the Initiation codon and the 3′ end of the 16S ribosomal RNA.

shoot apical meristem The major meristematic tissue of the plant; surrounds the shoot.

short interspersed elements (SINEs) Sequences of DNA interspersed in eukaryotic chromosomes in many copies. Alu, a three-hundred base-pair sequence, is found about half a million times in human DNA.

shotgun cloning The random cloning of pieces of the DNA of an organism without regard to the genes or sequences present in the cloned DNA.

shunting Process in which the first initiation codon on a messenger RNA is bypassed for an initiation codon further down the messenger. The process is probably guided by secondary structure in the messenger.

siblings (sibs) Brothers and sisters.

sigma factor The protein that gives promoter-recognition specificity to the RNA polymerase core enzyme of bacteria.

signal hypothesis The major mechanism whereby proteins that must insert into or across a membrane are synthesized by a membrane-bound ribosome. The first thirteen to thirty-six amino acids synthesized, termed a *signal peptide*, are recognized by a *signal recognition particle* that draws the ribosome to the membrane surface. The signal peptide may be removed later from the protein.

signal peptide *See* signal hypothesis.

signal recognition particle *See* signal hypothesis.

signal transduction pathway A pathway in which the action of kinase enzymes that free transcription factors, or some other action, translates an environmental signal into some form of gene action.

single-nucleotide polymorphisms (SNPs) Differences between individuals involving single base pairs that are located about every 1,000 bases along the human genome. SNPs are useful for mapping disease genes.

single-strand binding proteins Proteins that attach to single-stranded DNA, usually near the replicating Y-junction, to stabilize the single strands.

sister chromatids *See* chromatids.

site-specific recombination A crossover event, such as the integration of phage λ, that requires homology of only a very short region and uses an enzyme specific for that recombination.

small nuclear ribonucleoproteins (snRNPs) Components of the spliceosome, the intron-removing apparatus in eukaryotic nuclei. *See* snRNPs.

small nucleolar ribonucleoprotein particles (snoRNPs) Particles composed of RNA and protein found in the nucleolus that modify ribosomal RNAs, particularly by converting some uridines to pseudouridines and methylating some ribose sugars.

small nucleolar RNAs (snoRNAs) RNAs found in small nucleolar ribonucleoprotein particles (snoRNPs) that take part in modifying ribosomal RNA in the nucleolus.

SMC proteins For *s*tructural *m*aintenance of *c*hromosomes; proteins that aid mitotic segregation, sister-chromatid adhesion, dosage compensation, recombination, and other chromosomal activities.

snRNPs *See* small nuclear ribonucleoproteins.

sociobiology The study of the evolution of social behavior in animals.

somatic doubling A disruption of the mitotic process that produces a cell with twice the normal chromosome number.

somatic hypermutation The occurrence of a high level of mutation in the variable regions of immunoglobulin genes.

SOS box The region of the promoters of various genes that the LexA repressor recognizes. Release of repression results in the induction of the SOS response.

SOS response Repair systems (RecA, Uvr) induced by the presence of single-stranded DNA that usually occurs from postreplicative gaps caused by various types of DNA damage. The RecA protein, stimulated by single-stranded DNA, is involved in the inactivation of the LexA repressor, thereby inducing the response.

Southern blotting A method, first devised by E. M. Southern, used to transfer DNA fragments from an agarose gel to a nitrocellulose gel for the purpose of DNA-DNA or DNA-RNA hybridization during recombinant DNA work.

specialized transduction Form of transduction based on faulty looping out by a temperate phage. Only neighboring loci to the attachment site can be transduced. *See* generalized transduction.

speciation A process whereby, over time, one species evolves into a different species (anagenesis) or one species diverges to become two or more species (cladogenesis).

species A group of organisms capable of interbreeding to produce fertile offspring.

specific transcription factors Proteins needed for activation of transcription at specific eukaryotic promoters. Also, negative factors that can inhibit transcription at a specific eukaryotic promoter.

spermatids The four products of meiosis in males that develop into sperm.

spermatogenesis The process of sperm production.

spermatogonium A cell type in the testes of male vertebrates that gives rise to primary spermatocytes by mitosis.

sperm cells The gametes of males.

spermiogenesis The process by which spermatids mature into sperm cells.

spindle The microtubule apparatus that controls chromosomal movement during mitosis and meiosis.

spindle pole body Spindle microtubule organizing center found in fungi.

spiral cleavage The cleavage process in mollusks and some other invertebrates whereby the spindle is tipped at mitosis in relation to the original egg axis.

spirillum A spiral bacterium.

spliceosome Protein-RNA complex that removes introns in eukaryotic nuclear RNAs.

sporophyte The stage of a plant life cycle that produces spores by meiosis and alternates with the gametophyte stage.

stabilizing selection A type of selection that removes individuals from both ends of a phenotypic distribution, thus maintaining the same distributional mean.

standard deviation The square root of the variance.

standard error of the mean The standard deviation divided by the square root of the sample size. It is the standard deviation of a sample of means.

statistics Measurements of attributes of a sample from a population; denoted by Roman letters. *See* parameters.

stem-loop structure A lollipop-shaped structure formed when a single-stranded nucleic acid molecule loops back on itself to form a complementary double helix (stem), topped by a loop.

stepladder gel A DNA-sequencing gel. The numerous bands in each lane give the appearance of a stepladder.

stochastic A process with an indeterminate or random element as opposed to a deterministic process with no random element.

stringent factor A protein that catalyzes the formation of an unusual nucleotide (guanosine tetraphosphate) during the stringent response under amino acid starvation.

stringent response A translational control mechanism in prokaryotes that represses transfer RNA and ribosomal RNA synthesis during amino acid starvation.

structural alleles Alleles whose alterations include some of the same base pairs.

submetacentric chromosome A chromosome whose centromere lies between its middle and its end, but closer to the middle.

subtelocentric chromosome A chromosome whose centromere lies between its middle and its end, but closer to the end.

sum rule The rule that states that the probability that two mutually exclusive events will occur is the sum of the probabilities of the individual events.

supercoiling Negative or positive coiling of double-stranded DNA that differs from the relaxed state.

supergenes Several loci, which usually control related aspects of the phenotype, in close physical association.

suppressor gene A gene that, when mutated, apparently restores the wild-type phenotype to a mutant of another locus.

surveillance mechanism Used to describe mechanisms that oversee checkpoints in the cell cycle, points in which the process can be halted if certain conditions are not met.

survival of the fittest In evolutionary theory, survival of only those organisms best able to obtain and utilize resources (the fittest). This phenomenon is the cornerstone of Darwin's theory of evolution.

Svedberg unit A unit of sedimentation during centrifugation. Abbreviation is S, as in 50S.

swivelase *See* DNA gyrase.

sympatric speciation Speciation in which reproductive isolating mechanisms evolve within the range and habitat of the parent species. This type of speciation may be common in parasites.

synapsis The point-by-point pairing of homologous chromosomes during zygonema or in certain dipteran tissues that undergo endomitosis.

synaptonemal complex A proteinaceous complex that apparently mediates synapsis during the zygotene stage and then disintegrates.

syncitium A cell that has many nuclei not separated by cell membranes.

synexpression group A group of eukaryotic genes that are involved in the same function or pathway and are induced together.

synteny test A test that determines whether two loci belong to the same linkage group by observing concordance in hybrid cell lines.

synthetic medium A chemically defined substrate that microorganisms are grown upon.

TACTAAC box A consensus sequence surrounding the lariat branch point of eukaryotic messenger RNA introns.

TATA-binding protein A protein, part of TFIID, that binds the TATA consensus sequence in eukaryotic promoters.

TATA box An invariant DNA sequence at about −25 in the promoter region of eukaryotic genes; analogous to the Pribnow box in prokaryotes.

tautomeric shift Reversible shifts of proton positions in a molecule. The bases in nucleic acids shift between the keto and enol forms or between the amino and imino forms.

TBP-associated factors (TAFs) Proteins that bind with the TATA-binding protein to form TFIID. They aid in the selectivity of TFIID.

T-cell receptors Surface proteins of T cells that allow the T cells to recognize host cells that have been infected.

telocentric chromosome A chromosome whose centromere lies at one of its ends.

telomerase An enzyme that adds telomeric sequences to the ends of eukaryotic chromosomes.

telomeres The ends of linear chromosomes.

telophase The terminal stage of mitosis or meiosis in which chromosomes uncoil, the spindle breaks down, and cytokinesis usually occurs.

telson The posterior end of the arthropod embryo, where the end of the alimentary canal is located.

temperate phage A phage that can enter into lysogeny with its host.

temperature-sensitive mutant An organism with an allele that is normal at a permissive temperature but mutant at a restrictive temperature.

template A pattern serving as a mechanical guide. In DNA replication, each strand of the duplex acts as a template for the synthesis of a new double helix.

template strand *See* anticoding strand.

terminator sequence A sequence in DNA that signals the termination of transcription to RNA polymerase.

terminator stem A configuration of the leader transcript that signals transcriptional termination in attenuator-controlled amino acid operons.

tertiary structure The further folding of a protein, bringing α helices and β sheets into three-dimensional arrangements.

testcross The cross of an organism with a homozygous recessive organism.

testing of hypotheses The determination of whether to reject or fail to reject a proposed hypothesis based on the likelihood of the experimental results.

testis-determining factor (*TDF*) General term for the gene determining maleness in human beings (*Tdf* in mice).

tetrads The meiotic configuration of four chromatids first seen in pachynema. There is one tetrad (bivalent) per homologous pair of chromosomes.

tetranucleotide hypothesis The hypothesis, based on incorrect information, that DNA could not be the genetic material because its structure was too simple—that is, that repeating subunits contain one copy each of the four DNA nucleotides.

tetraploids Organisms with four whole sets of chromosomes.

tetratype (TT) A spore arrangement in Ascomycete fungi that consists of the two parental and two recombinant spores.

theta structure An intermediate structure formed during the replication of a circular DNA molecule.

three-point cross A cross involving three loci.

thymine *See* pyrimidines.

t-loop A loop that forms at the end of mammalian telomeres by the interdigitation of the 3′ free end into the DNA double helix.

topoisomerase An enzyme that can relieve (or create) supercoiling in DNA by creating transitory breaks in one (type I) or both (type II) strands of the helical backbone.

topoisomers Forms of DNA with the same sequence but differing in their linkage number (coiling).

totipotent The state of a cell that can give rise to any and all adult cell types, as compared with a differentiated cell whose fate is determined.

trailer The length of messenger RNA from the nonsense codon to the 3′ end or, in polycistronic messenger RNAs, from a nonsense codon to the next gene's leader.

trans Meaning "across" and referring usually to the geometric configuration of mutant alleles across from each other on a homologous pair of chromosomes.

***trans*-acting** Referring to mutations of, for example, a repressor gene, that act through a diffusible protein product; the normal mode of action of most recessive mutations.

transcription The process whereby RNA is synthesized from a DNA template.

transcription factors Eukaryotic proteins that aid RNA polymerase in recognizing promoters. *See* general transcription factors *and* specific transcription factors.

transducing particle A defective phage, carrying part of the host's genome.

transduction A process whereby a cell can gain access to and incorporate foreign DNA brought in by a viral particle.

transfection The introduction of foreign DNA into eukaryotic cells.

transfer operon (*tra*) Sequence of loci that impart the male (F-pili-producing) phenotype on a bacterium. The male cell can transfer the F plasmid to an F^- cell.

transfer RNA (tRNA) Small RNA molecules that carry amino acids to the ribosome for polymerization.

transformation A process whereby prokaryotes take up DNA from the environment and incorporate it into their genomes, or the conversion of a eukaryotic cell into a cancerous one.

transgenic Eukaryotic organisms that have taken up foreign DNA.

transition mutation A mutation in which a purine-pyrimidine base pair replaces a base pair in the same purine-pyrimidine relationship.

translation The process of protein synthesis wherein the nucleotide sequence in messenger RNA determines the primary structure of the protein.

translocase (EF-G) Elongation factor in prokaryotes necessary for proper translocation at the ribosome during the translation process. Replaced by eEF2 in eukaryotes.

translocation A chromosomal configuration in which part of a chromosome becomes attached to a different chromosome. Also a part of the translation process in which the messenger RNA is shifted one codon in relation to the ribosome.

translocation channel (translocon) A protein-lined pore or channel in a membrane through which nascent proteins are transported during translation.

transposable genetic element A region of the genome, flanked by inverted repeats, a copy of which can be inserted at another place; also called a transposon or a jumping gene.

transposon *See* transposable genetic element.

transversion mutation A mutation in which a purine replaces a pyrimidine, or vice versa.

trihybrid An organism heterozygous at three loci.

triploids Organisms with three whole sets of chromosomes.

trisomic A diploid cell with an extra chromosome.

tRNA *See* transfer RNA (tRNA).

trp **RNA-binding attenuation protein (TRAP)** Protein that can bind to the attenuation region of the messenger RNA of the tryptophan operon in *Bacillus subtilis,* causing a terminator stem to form and halting further transcription.

true heritability *See* heritability.

tumor Abnormal growth of tissue.

tumor-suppressor genes Genes that normally prevent unlimited cellular growth. When both copies of the gene are mutated, cellular transformation follows. Examples are the *p53* gene and the genes for retinoblastoma and Wilm's tumor.

two-point cross A cross involving two loci.

type I error In statistics, the rejection of a true hypothesis.

type II error In statistics, the accepting of a false hypothesis.

typological thinking The concept that organisms of a species conform to a specific norm. In this view, variation is considered abnormal.

ubiquitin A peptide of twenty-six amino acid residues that enzymes attach to proteins that the proteasome will degrade.

unequal crossing over Nonreciprocal crossing over caused by the mismatching of homologous chromosomes. Usually occurs in regions of tandem repeats.

uninemic chromosome A chromosome consisting of one DNA double helix.

unique DNA A length of DNA with no repetitive nucleotide sequences.

unmixed families Groups of four codons sharing their first two bases and coding for the same amino acid.

unusual bases Other bases, in addition to adenine, cytosine, guanine, and uracil, found primarily in transfer RNAs.

UP element *See* upstream element.

upstream A convention on DNA related to the position and direction of transcription by RNA polymerase ($5' \rightarrow 3'$). Downstream (or $3'$ to) is in the direction of transcription whereas upstream ($5'$ to) is in the direction from which the polymerase has come.

upstream element A sequence of about twenty AT-rich bases centered at -50 in promoters of prokaryotic genes that are expressed strongly.

uracil *See* pyrimidines.

variable-number-of-tandem-repeats (VNTR) loci Loci that are hypervariable because of tandem repeats. Presumably, variability is generated by unequal crossing over.

variance The average squared deviation about the mean of a set of data.

variegation Patchiness; a type of position effect that results when particular loci are contiguous with heterochromatin.

virion A virus particle.

viroids Bare RNA particles that are plant pathogens.

V(D)J joining The process of joining variable, diversity, and joining gene segments (V-J, V-D, or D-J joining) in the formation of a functioning immunoglobulin gene.

Wahlund effect The effect of subdivision on a population, causing it to contain fewer heterozygotes than predicted despite the fact that all subdivisions are in Hardy-Weinberg proportions.

western blotting A technique for probing for a particular protein using antibodies. *See* Southern blotting.

whole-genome shotgun method A method of sequencing entire genomes by breaking up the genomes into small pieces, sequencing the pieces, and then using computers to establish order by overlapping the sequences.

wild-type The phenotype of a particular organism when it is first seen in nature.

Wilm's tumor A childhood kidney cancer caused by the inactivation of an anti-oncogene.

wobble Referring to the reduced constraint over the third base of an anticodon as compared with the other bases, thus allowing additional complementary base pairings.

xeroderma pigmentosum A disease in human beings caused by a defect in the UV mutation repair system.

X-inactivation center (*XIC*) Locus at which inactivation is initiated on the X chromosome in mammals.

X linked *See* sex-linked.

X-ray crystallography A technique, using X rays, to determine the atomic structure of molecules that have been crystallized.

yeast artificial chromosome (YAC) Originating from a bacterial plasmid, a YAC contains additionally a yeast centromeric region (CEN) and a yeast origin of DNA replication (ARS). YACs are capable of including large pieces of foreign DNA during cloning.

Y-junction The point of active DNA replication where the double helix opens up so that each strand can serve as a template.

Y linked Inheritance pattern of loci located on the Y chromosome only. Also refers to the loci themselves.

Z DNA A left-handed form of DNA found under physiological conditions in short GC segments that are methylated. It may be involved in regulating gene expression in eukaryotes.

ZFY **gene** Originally believed to be the human male sex-switch gene, located on the short arm of the Y chromosome. *ZFY* stands for *z*inc *f*inger on the *Y* chromosome.

zinc finger Configuration of a DNA-binding protein that resembles a finger with a base, usually cysteines and histidines, binding a zinc ion. Discovered in a transcription factor in *Xenopus.*

zygonema (zygotene stage) The stage of prophase I of meiosis in which synapsis occurs.

zygotic induction The beginning of vegetative growth when a prophage is passed into an F^- cell during conjugation.

zygotic selection The forces acting to cause differential mortality of an organism at any stage (other than gametes) in its life cycle.

INDEX

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